

Project Report

Semester: 4

ME2171

Reverse Engineering of a Hand Mixer

By

Group 03

Date of submission	13/05/2026
Due date of submission	13/05/2026

**Department of Mechanical Engineering
University of Moratuwa
Sri Lanka**

Table of Contents

Table of Contents.....	2
Individual Contribution.....	5
1. Housing of the hand mixer.....	6
1.1 Functions and features of the hand mixer	6
1.2 Component Analysis.	6
1.2.1 Dimensions	6
1.2.2 Material Identification Process	8
1.3 Manufacturing processes.....	15
Possible manufacturing process for the housing.....	15
Reasons for selecting injection moulding	15
Injection moulding for housing.....	16
Surface Finish	18
Changes in material due to injection moulding	18
1.4 Drawings and 3d assembly of the housing.....	19
1.5 Improvements to the housing	21
2 Cooling fan and gears	22
2.1 Functions	22
2.2 Product analysis.....	22
2.2.1. Possible tests for polymers.....	22
2.2.2. Conducted tests for polymers.....	23
2.2.3 Test conclusions.....	25
2.3 Improved material choice for gear manufacturing	28

2.4	3D model and 2D drawings.....	28
2.5	Manufacturing methods.....	31
2.5.1	Possible Manufacturing methods.....	31
2.5.2	Why injection moulding?.....	31
2.5.3	Why CNC is not a suitable for gear manufacturing?	32
2.5.4	Injection moulding	33
2.5.5	Surface finish	35
3	Beater of the hand mixer.....	36
3.1	Functions of the Beater	36
3.2	Material identification process	36
3.3	Manufacturing methods.....	39
3.3.1	Process of CNC wire forming of a beater.....	39
3.3.2	Finishing Processes.....	40
3.4	2D Drawing and 3D Modelling of the beater.....	41
3.5	Improvements suggested for the beater.....	43
4	Motor Housing	44
4.1	Functional properties of the motor housing in hand mixer.	44
4.2	Product Analysis of Motor Housing.....	45
	Density test.....	45
	Spark Test	47
	Spark-Optical Emission Spectrometry (OES)	48
4.3	Manufacturing Methods	49
4.4	Measurement Process	54

4.5	3D CAD model.....	54
4.6	Drawing.....	55
5	Hand Mixer Motor Shaft.....	56
5.1	Functions and features of the hand mixer	56
5.2	Material Identification.....	57
	Density Test	57
	Spark Test	57
5.3	Identified Manufacturing Methods.....	59
5.4	5.4 3d model and drawing.....	60
5.5	Proposed Improvements	61
6	Assembly.....	63
6.1	Assembling the components of the hand mixer	63
6.2	3D Assembly exploded view.....	63
6.3	Final assembly.....	64
7	References.....	65

Individual Contribution

Name	Index Number	Respective component	Contribution
JANURAJAH J.	230264R	Motor Shaft	20%
MAKULDENIYA M.W.M.D.U.G.	230388B	Gears and Cooling fan	20%
SANJAYA G.H.D.N.	230576B	Motor Housing	20%
VISHUDHDHA P.G.T.	230669M	Beaters	20%
SENADHEERA H.W.M.L.U.C.	230593A	Housing	20%

1. Housing of the hand mixer

1.1 Functions and features of the hand mixer

- Holds all the components inside the hand mixer - motor, motor housing, cooling fan, gears, beaters.
- Consists of two major parts with top and bottom of housing.
- Designed to use with one hand.
- Design is symmetric; both right-handed and left-handed people can use.
- Inside of the hand mixer is well ventilated.

1.2 Component Analysis.

1.2.1 Dimensions

The dimensions were taken using vernier caliper, micrometer screw gauge, height gauge.



Figure 1



Figure 2

The curvature of the outer surfaces was taken by sectioning the body at specific distances, placing a thread over the part at that section and the length of the thread was measured to get the idea of size of curvature.

Following are other accurate methods to obtain the curvatures of bodies.

- Radius measuring gauges



Figure 3



Figure 5 - Setting the gauge to zero



Figure 4 - Measuring the radius of the component

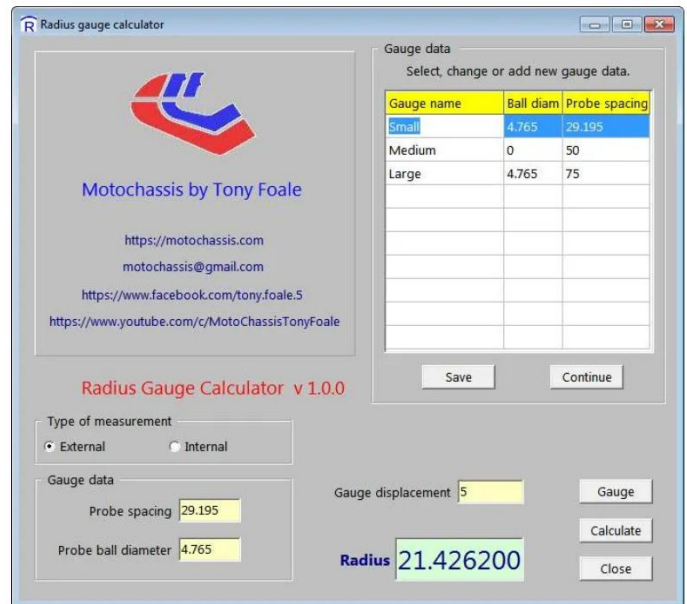
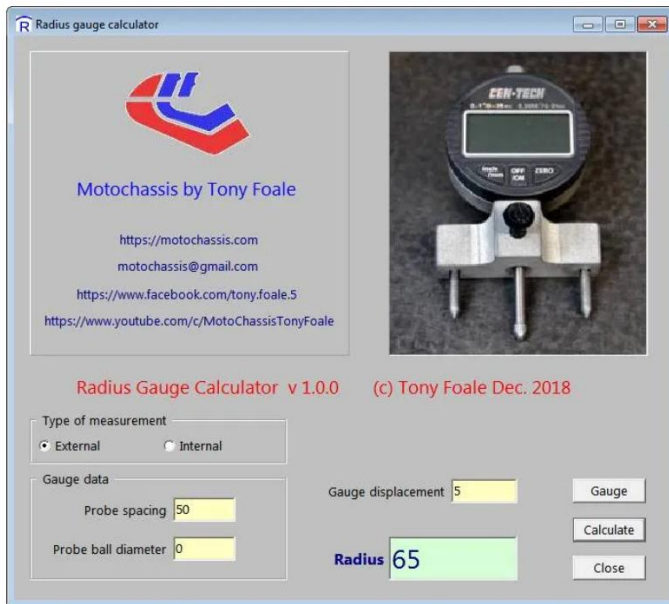


Figure 6 - Enter the gauge reading, probe diameter, probe spacing in the software and obtain the relevant magnitude radius measured.

- Optical 3D Scanning
 - Structured light scanning
 - Lase triangulation scanning
 - Photogrammetry
- Contact 3D Scanning
 - Coordinate Measuring Machine (CMM)

1.2.2 Material Identification Process

1. Visual Inspection

Objective	Observations and Conclusions
To observe the surface texture, hardness/flexibility, and appearance of the component	Both parts were manufactured by plastic (polymer) material. Have high-quality surface and gloss. Hard Opaque

2. Float Test

Objective	Procedure	Observations	Conclusion
To obtain qualitative comparison of the specific density of the materials of top and bottom parts of the housing	Two samples of each part were put into a water container.	Both the samples sank in water.	Density of both materials is higher than the density of water.

3. Shore D Hardness test

Objective	Procedure	Observations	Conclusion
To get measurements for the hardness of the material of the component. To identify the material by comparing with	Test – Shore D hardness Test. Instrument – Durometer – GS702-Type D (Figure 8)	Readings (Top Part) - 78,78,79,80 Readings (Bottom Part) - 62, 59,64,65	Compared the readings with the Shore D hardness scale (Figure 11). Hardness values of both parts are in extra hard region.

hardness vales of polymers.			
-----------------------------	--	--	--

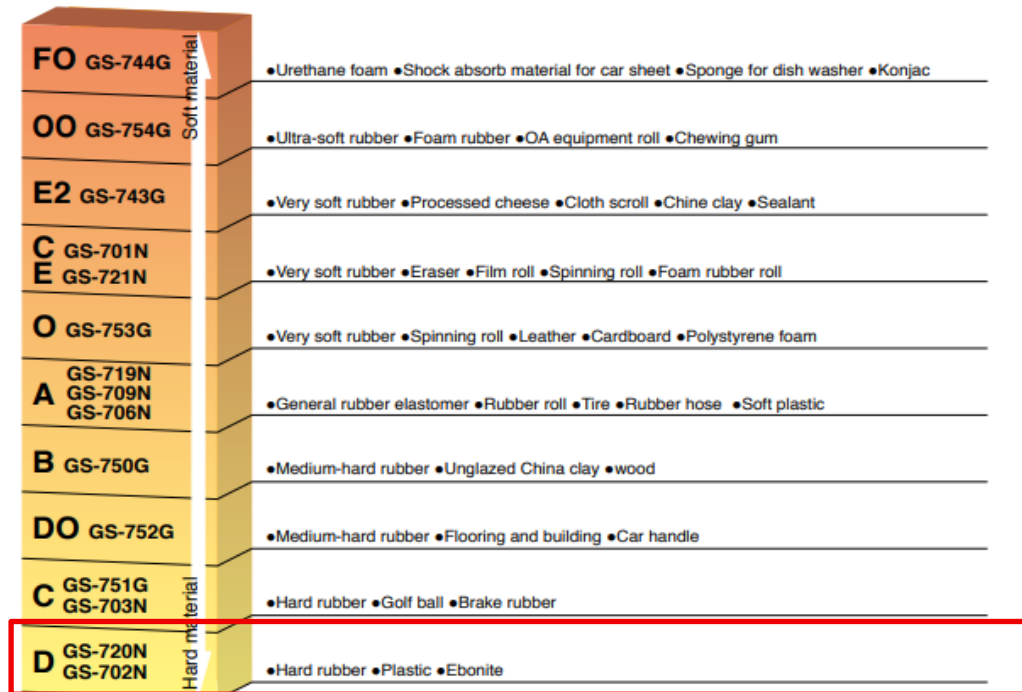


Figure 7 - Durometer Categorization according to hardness values and materials



Figure 8 - GS - 702N - Type D Durometer

Top Part



Figure 9

Bottom Part



Figure 10

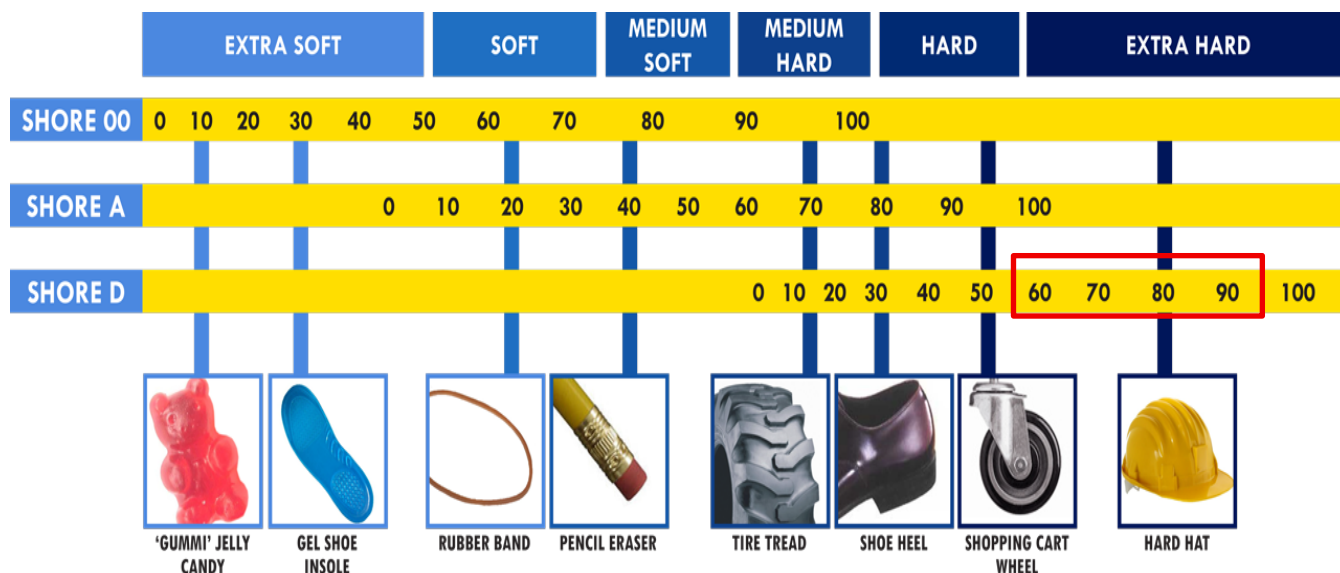


Figure 11- Smooth on - Durometer Shore Hardness Scale

4. Density Test

Objective	Procedure	Observations	Conclusion
To compare the densities quantitatively with the known ranges of densities of polymers.	1. Measuring Density by pycnometer (Figure 12)	Top part - 1.068 g/cm^3 Bottom part - 1.092 g/cm^3 (Figure 13,14)	Compared the density values and hardness vales of two parts with MatWeb property data (https://www.matweb.com/search/PropertySearch.aspx). Material of both parts - Acrylonitrile Butadiene Styrene (ABS) (Figure 16,17)
	2. Measuring Density by Density meter (Figure 15)	Top part - 0.994 g/cm^3 Bottom part - 0.984 g/cm^3	



Pycnometer



Pycnometer - <https://www.labworld.co.uk/gay-lussac-pycnometer.html>

Figure 12

Top part of the housing



Measuring mass of pycnometer



Measuring mass of pycnometer + sample



Measuring mass of pycnometer + water



Measuring mass of pycnometer + water + sample

Figure 13

Bottom part of the housing



Measuring mass of pycnometer + sample
Top



Measuring mass of pycnometer + water +
Bottom part

Figure 14

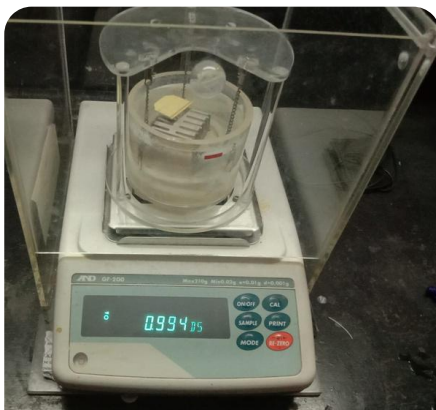


Figure 15

Overview of materials for Acrylonitrile Butadiene Styrene (ABS), Molded

Categories: [Polymer](#); [Thermoplastic](#); [ABS Polymer](#); [Acrylonitrile Butadiene Styrene \(ABS\)](#); [Molded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Acrylonitrile Butadiene Styrene (ABS), Molded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

[Printer friendly version](#) [Download as PDF](#) [Download to Excel \(requires Excel and Windows\)](#)
[Export data to your CAD/FEA program](#)

Add to Folder: ☐ My Folder 0/0

Physical Properties	Metric	English	Comments
Density	0.882 - 3.50 g/cc	0.0319 - 0.126 lb/in ³	Average value: 1.07 g/cc Grade Count:774
Water Absorption	0.0250 - 2.30 %	0.0250 - 2.30 %	Average value: 0.383 % Grade Count:85
Moisture Absorption at Equilibrium	0.100 - 0.300 %	0.100 - 0.300 %	Average value: 0.232 % Grade Count:26
Water Absorption at Saturation	0.00950 - 1.03 %	0.00950 - 1.03 %	Average value: 0.561 % Grade Count:14
Maximum Moisture Content	0.0100 - 0.150	0.0100 - 0.150	Average value: 0.0459 Grade Count:71
Linear Mold Shrinkage	0.000 - 0.0290 cm/cm	0.000 - 0.0290 in/in	Average value: 0.00533 cm/cm Grade Count:581
Linear Mold Shrinkage, Transverse	0.00200 - 0.00900 cm/cm	0.00200 - 0.00900 in/in	Average value: 0.00522 cm/cm Grade Count:52
Melt Flow	0.0800 - 125 g/10 min	0.0800 - 125 g/10 min	Average value: 16.5 g/10 min Grade Count:805
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell M	53.0 - 92.0	53.0 - 92.0	Average value: 71.8 Grade Count:4
Hardness, Rockwell R	13.0 - 122	13.0 - 122	Average value: 108 Grade Count:431
Hardness, Shore D	79.0 - 103	79.0 - 103	Average value: 87.0 Grade Count:3

Figure 16 – For Top part

Overview of materials for Acrylonitrile Butadiene Styrene (ABS)/Nylon Blend

Categories: [Polymer](#), [Thermoplastic](#), [ABS Polymer](#), [Acrylonitrile Butadiene Styrene \(ABS\)/Nylon Blend](#), [Nylon \(Polyamide PA\)](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Acrylonitrile Butadiene Styrene (ABS)/Nylon Blend". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

[Printer friendly version](#) [Download as PDF](#) [Download to Excel \(requires Excel and Windows\)](#)
[Export data to your CAD/FEA program](#)

Add to Folder: ☐ My Folder 0/0

Physical Properties	Metric	English	Comments
Density	1.04 - 1.21 g/cc	0.0375 - 0.0437 lb/in ³	Average value: 1.08 g/cc Grade Count:100
	0.900 - 0.900 g/cc @Temperature 250 - 250 °C	0.0325 - 0.0325 lb/in ³ @Temperature 500 - 500 °F	Average value: 0.900 g/cc Grade Count:2
Water Content	8.00 - 9.00 %	8.00 - 9.00 %	Average value: 8.33 % Grade Count:3
Water Absorption	0.200 - 6.40 %	0.200 - 6.40 %	Average value: 2.77 % Grade Count:42
Moisture Absorption at Equilibrium	0.800 - 2.00 %	0.800 - 2.00 %	Average value: 1.43 % Grade Count:42
Water Absorption at Saturation	1.40 - 4.30 %	1.40 - 4.30 %	Average value: 2.48 % Grade Count:11
Maximum Moisture Content	0.0200 - 0.0500	0.0200 - 0.0500	Average value: 0.0400 Grade Count:3
Linear Mold Shrinkage	0.00400 - 0.0140 cm/cm	0.00400 - 0.0140 in/in	Average value: 0.00776 cm/cm Grade Count:64
	0.00700 - 0.00700 cm/cm @Temperature 50.0 - 50.0 °C	0.00700 - 0.00700 in/in @Temperature 176 - 176 °F	Average value: 0.00700 cm/cm Grade Count:1
Linear Mold Shrinkage, Transverse	0.00600 - 0.0140 cm/cm	0.00600 - 0.0140 in/in	Average value: 0.00839 cm/cm Grade Count:23
	0.00800 - 0.00800 cm/cm @Temperature 50.0 - 50.0 °C	0.00800 - 0.00800 in/in @Temperature 176 - 176 °F	Average value: 0.00800 cm/cm Grade Count:1
Melt Flow	1.00 - 65.0 g/10 min	1.00 - 65.0 g/10 min	Average value: 22.9 g/10 min Grade Count:57
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	69.0 - 116	69.0 - 116	Average value: 101 Grade Count:29
Hardness, Shore D	64.0	64.0	Average value: 64.0 Grade Count:3

Figure 17 – For bottom part

5. Burn Test

Objective	Procedure	Observations	Conclusion
To narrow down the polymer material by observing its behaviour under combustion	One sample piece of each part was burnt using a bunsen burner.	(Same observations were obtained for both samples.) - Continued burnt after removing from flame - Burnt at a moderate speed - Drips were observed. - Flame colour - bright yellow (orange) - Pungent smell - Black Smoke	By Plastic observation flow chart (Figure18,19); Material of both parts – ABS.

- By above conclusions of each test,
Material of top and the bottom parts - Acrylonitrile Butadiene Styrene (ABS).

PLASTICS IDENTIFICATION FLOW CHART

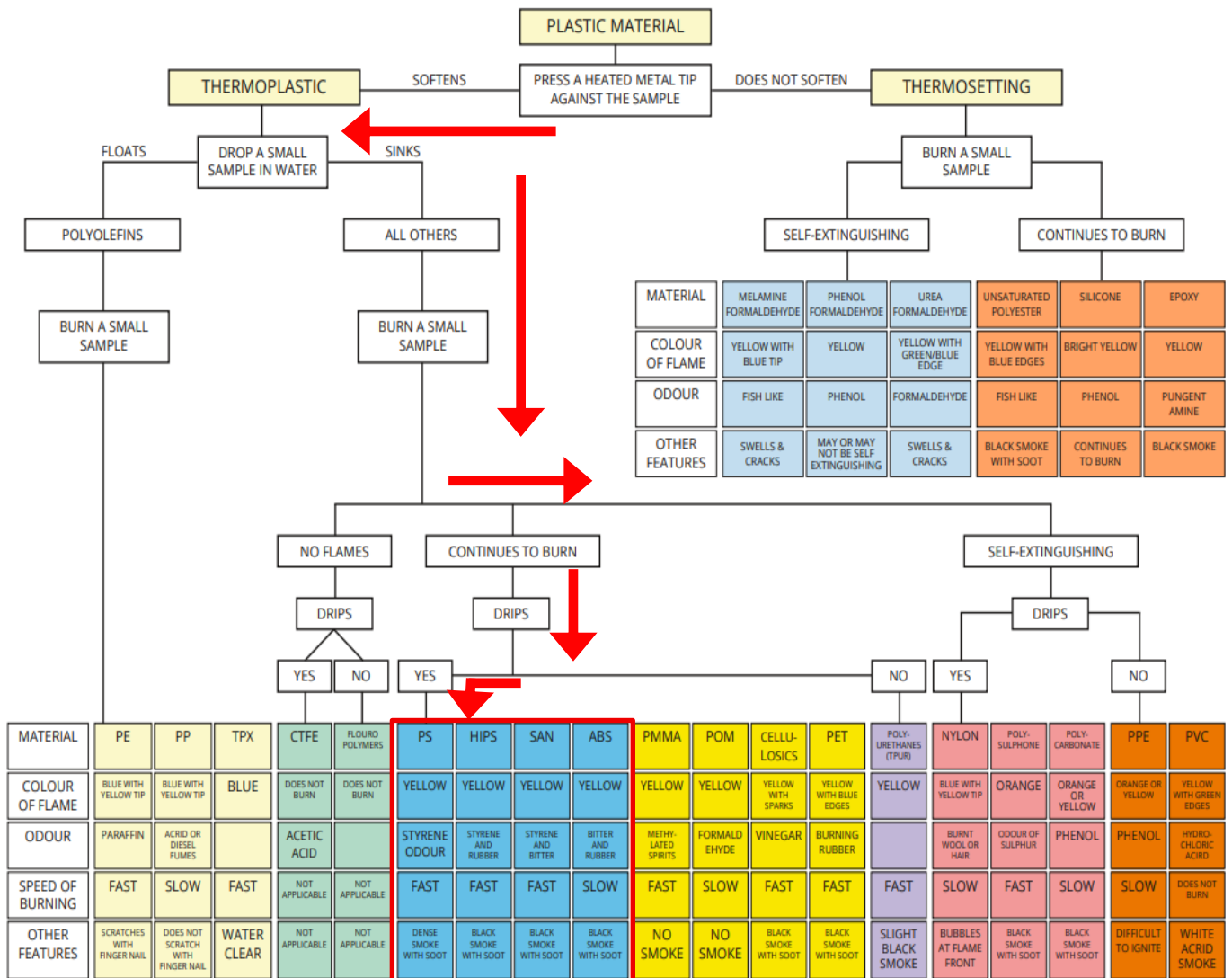


Figure 18 - Articles - STANMECH Technologies Inc

PS	HIPS	SAN	ABS
YELLOW	YELLOW	YELLOW	YELLOW
STYRENE ODOUR	STYRENE AND RUBBER	STYRENE AND BITTER	BITTER AND RUBBER
FAST	FAST	FAST	SLOW
DENSE SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT

Figure 19

1.3 Manufacturing processes

Possible manufacturing process for the housing

- For both parts - Injection moulding

Reasons for selecting injection moulding

- Suitable for high volume manufacturing processes with consistent shapes and complex geometries
- High precision
- Good surface quality
- Low cost per unit production
- Easy to add integrated features – screws, bosses, texts, etc.

Features visible in housing that justify the manufacturing process.

Top Part

- Parting line

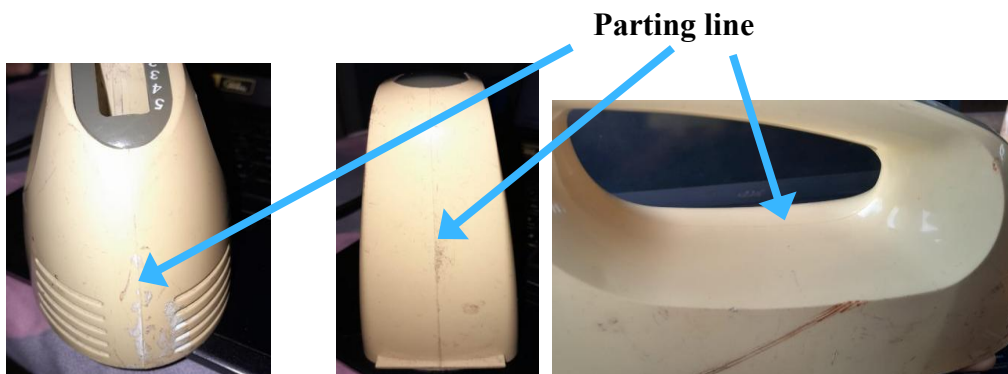


Figure 20

- Bosses

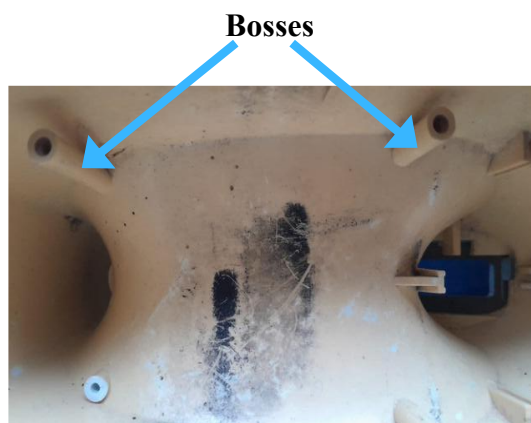


Figure 21

Bottom Part

- Ejector Pin marks
- Bosses

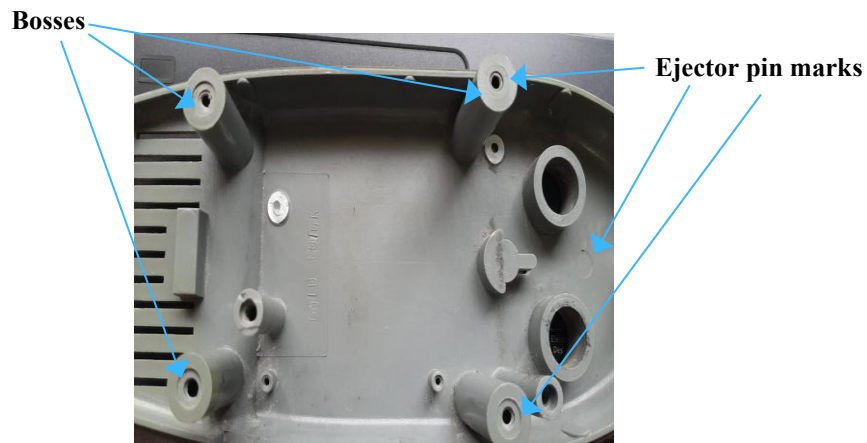


Figure 22

Injection moulding for housing

1. Mould design

- Product and mould design using CAD.
- Precise machining of mould by steel/aluminium.
- Features added to the mould.

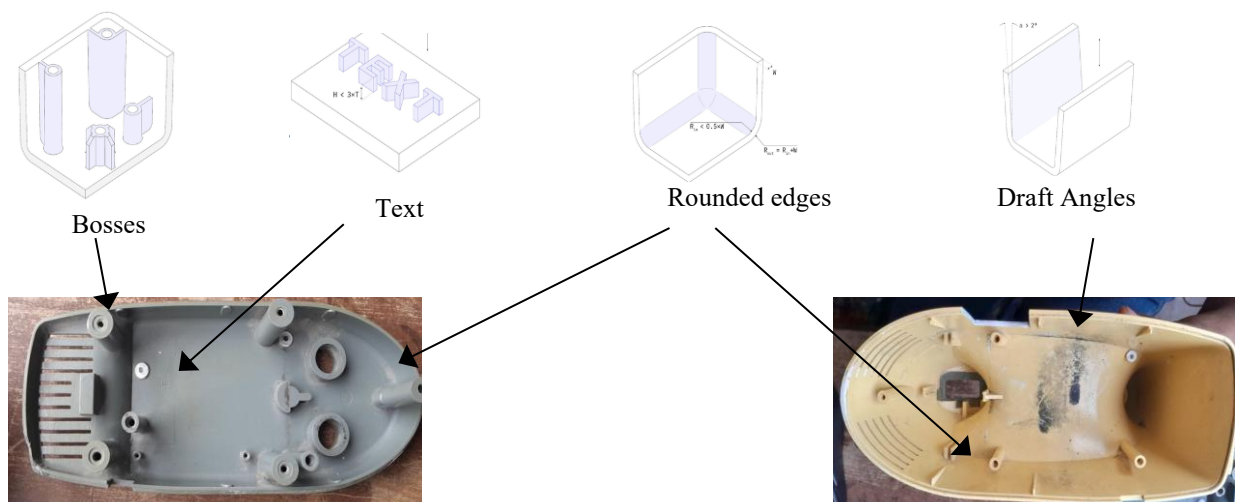


Figure 23

2. Material preparation

- ABS plastic pellets are melted and additives are added.

Material	Melting Point (°C)	Mould Temperature (°C)
ABS	200 - 250	50 - 80

3. Injection Moulding

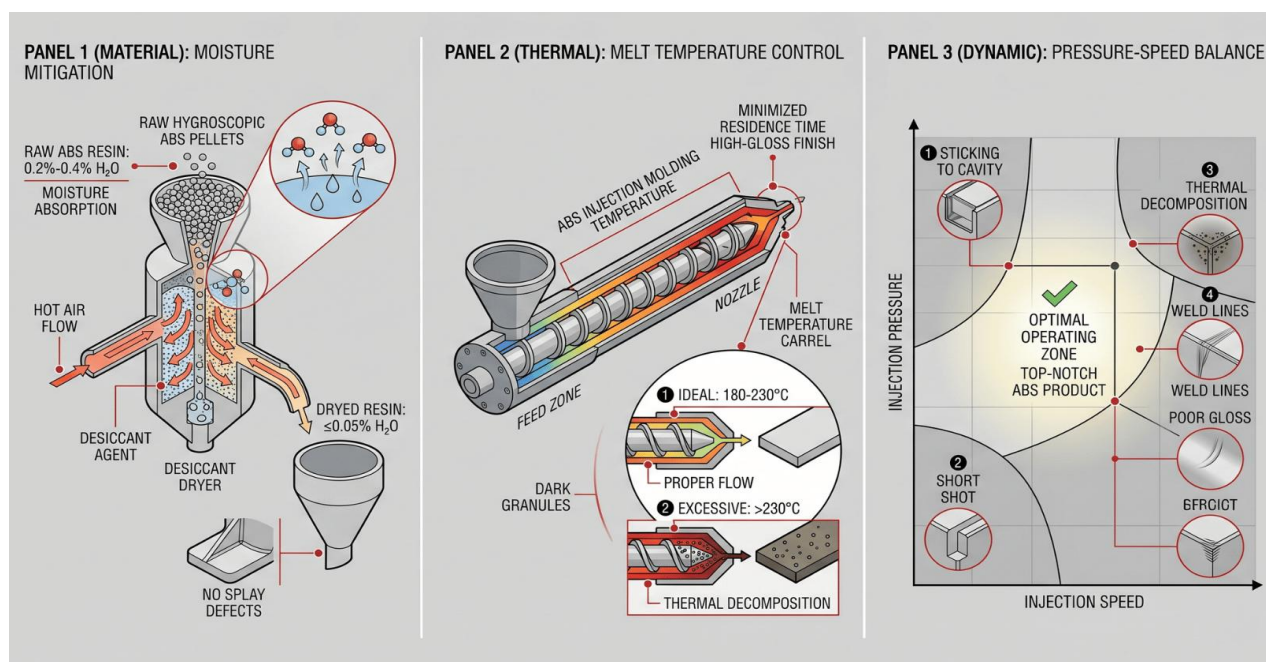
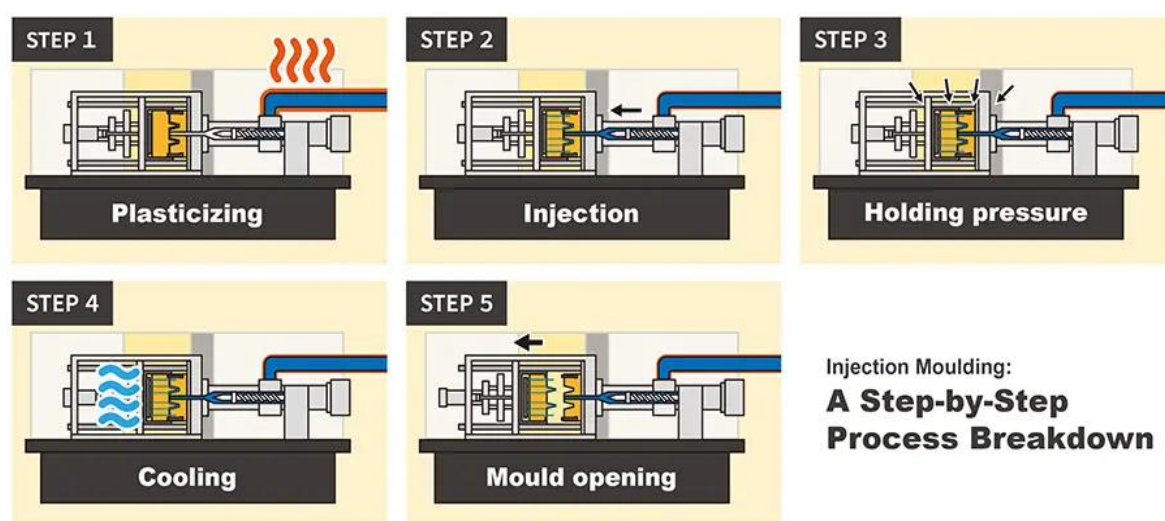


Figure 24 - <https://www.rapiddirect.com/blog/abs-injection-molding/>



<https://www.thy-precision.com>

THY HONG YANG PRECISION
PLASTIC INJECTION MOLDING SOLUTION

Figure 25

4. Self-tapping screws to create screw threads in bosses during assembly.

Surface Finish

- Top and bottom parts

Outer surface – Smooth, high gloss

Inner surface – Smooth, semi – gloss

Finish	Description	SPI standards*	Applications
Glossy finish	The mold is first smoothed and then polished with a diamond buff, resulting in a mirror-like finish.	A-1 A-2 A-3	Suitable for parts that require the smoothest surface finish for cosmetic or functional purposes (Ra less than 0.10 μm). The A-1 finish is suitable for parts with mirror-like finish and lenses.
Semi-gloss finish	The mold is smoothed with fine grit sandpaper, resulting in	B-1 B-2 B-3	Suitable for parts that require a good visual appearance, but not a

Figure 26 - Society of Plastic Industry (SPI) standards for surface finish

- By proper material selection

Material	Gloss Level	Common Applications
PS	High	Toys, containers
ABS	High	Electronics, auto parts

Figure 27 - <https://www.moldall.com/high-gloss-finish-injection-molding/>

- By controlling parameters in moulding process

Eg: temperature

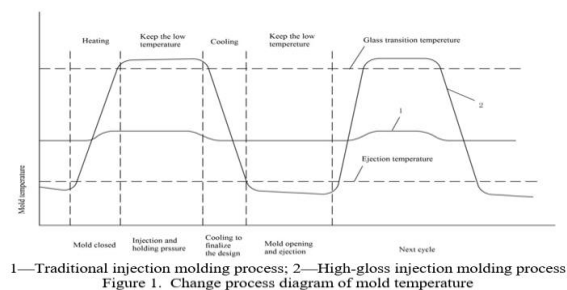


Figure 28- Development and Application of High-gloss Injection Molding Technology - Junjie Zhu et al 2022 J. Phys.: Conf. Ser.

2206 012016

- Improve directional strength (during filling of mould)

- Increase tensile stress (due to fast cooling)
- Variations in resistance to temperature
- Gloss appearance
- Volumetric shrinkage
- Reduce residual stress (by cooling)

1.4 Drawings and 3d assembly of the housing

Top part

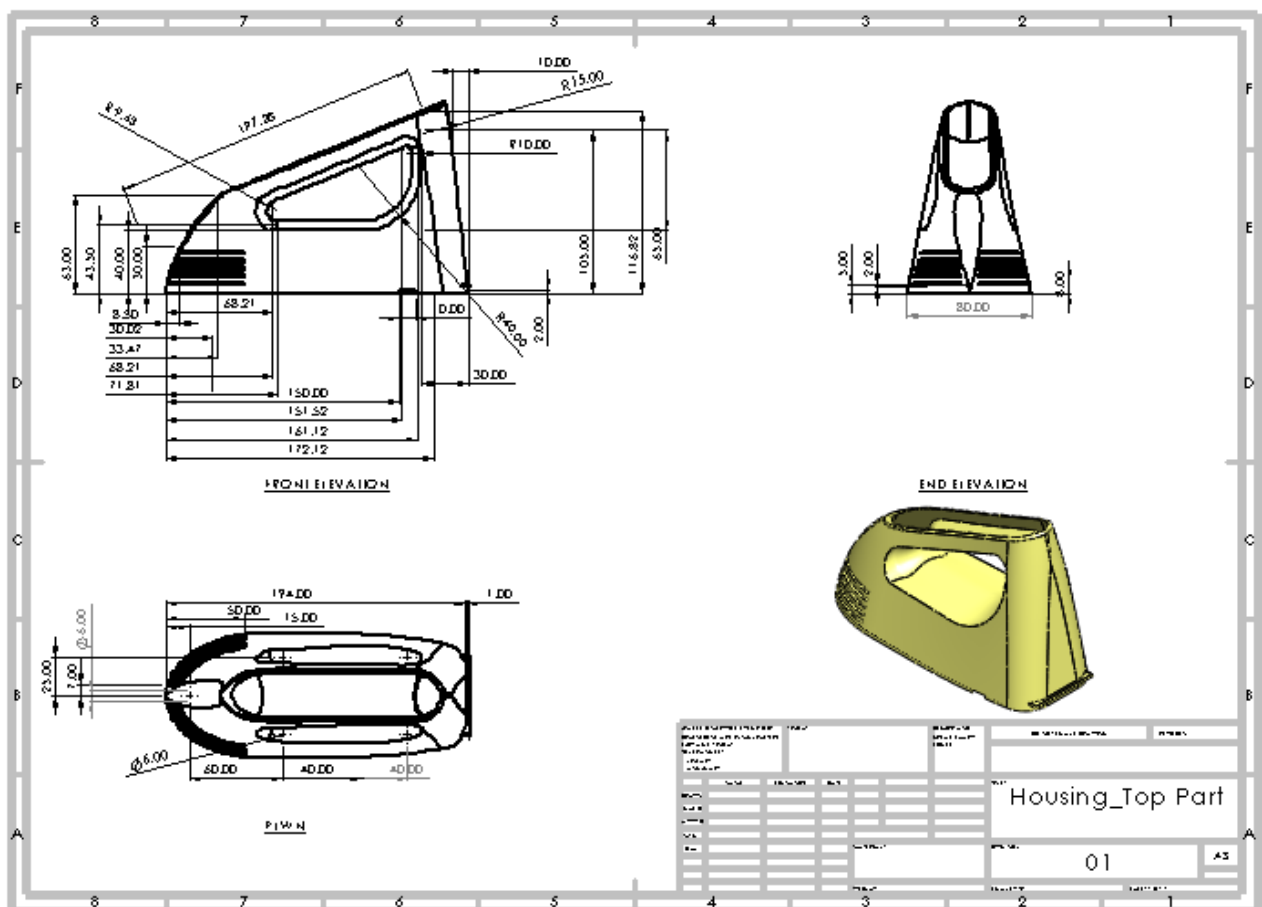


Figure 29

Bottom part

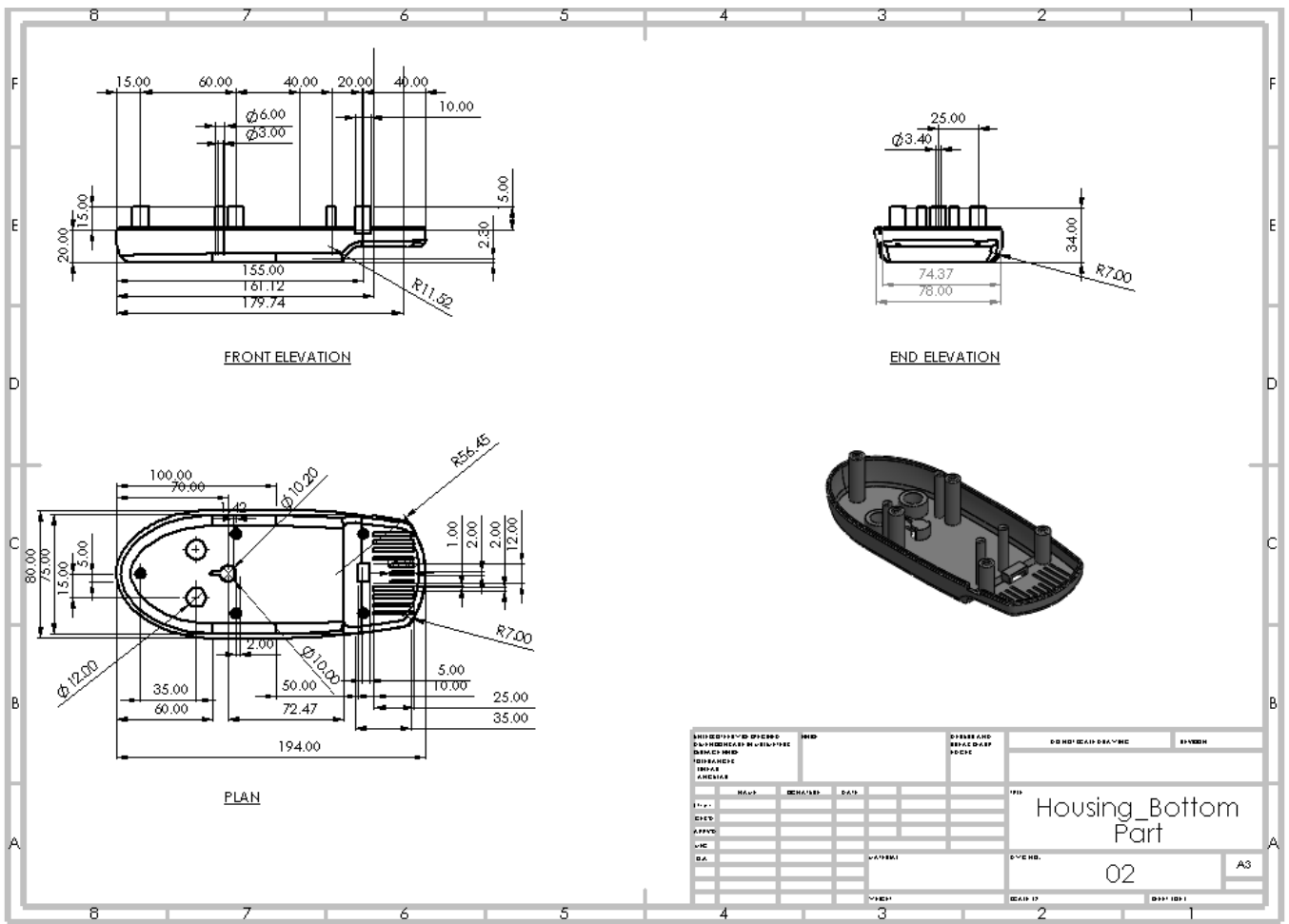
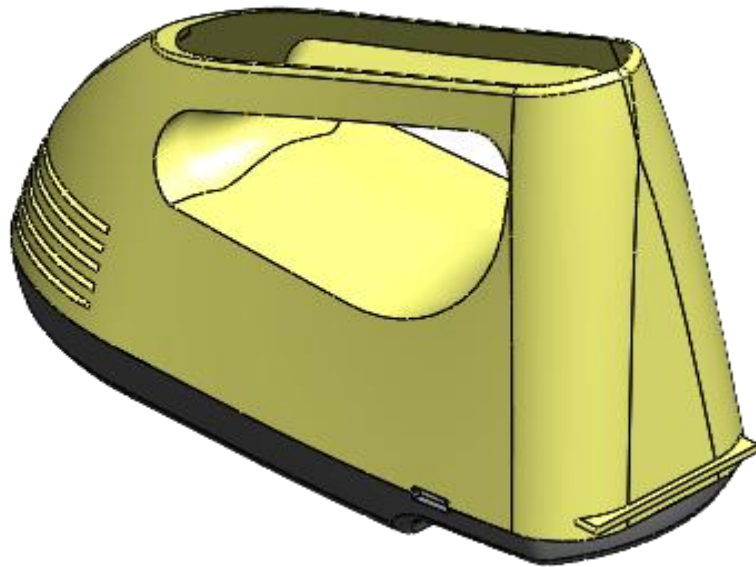


Figure 30

3D assembly of top and bottom parts



1.5 Improvements to the housing

Adding a small bent below the handle to improve,

- Ergonomics of handle
- Comfort for users



Figure 31- Original Design



Figure 32 – Improvement to handle

2 Cooling fan and gears

2.1 Functions

- Gear - transfer rotational energy to the beaters
- Cooling fan - Circulates air to remove heat and prevent overheating

2.2 Product analysis

2.2.1. Possible tests for polymers

Non - standard tests

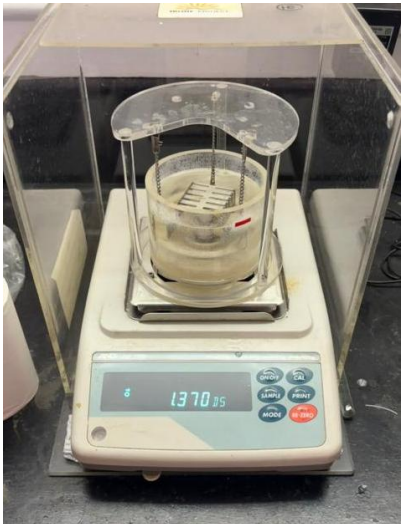
Visual inspection	Check colour, texture, flexibility, and transparency
Burn test	Observe combustion behaviour for polymer identification
Float test	Compare density relative to water

Standard tests

FTIR	Identifies chemical bonds using infrared absorption
Shore D Hardness (GS-702N)	Measures hardness of hard plastics and rubber
Density test (ISO 1183)	Measures material specific density
DSC	Measures thermal behaviour during heating

2.2.2. Conducted tests for polymers

Test	Cooling fan	Gear
Visual inspection	Smooth, flexible, translucent blades  <i>Figure 31 – cooling fan</i>	smooth, off-white, hard, opaque  <i>Figure 32 – gear</i>
Float test	Sinks in water	Sinks in water
Burn test	▪ Burn at moderate burn ▪ orange flame ▪ molten drips ▪ self-extinguishing ▪ pungent smell ▪ hard breakable residue  <i>Figure 33 – Burn test on cooling fan</i>	▪ Slow burn ▪ Blue base/yellow edge flame ▪ self-extinguishing ▪ pungent smell ▪ bubbles at flame front ▪ no smoke  <i>Figure 34 – Burn test on gear</i>

Shore D hardness test	73  <i>Figure 35 – Shore D value on cooling fan material</i>	72  <i>Figure 36 – Shore D value on gear material</i>
Density meter	1.098 g/c  <i>Figure 37 – Density meter value for cooling fan</i>	1.370 g/cm³  <i>Figure 38 – Density meter value for gear</i>
Pycnometer	1.22 g/cm³	1.47 g/cm³

2.2.3 Test conclusions

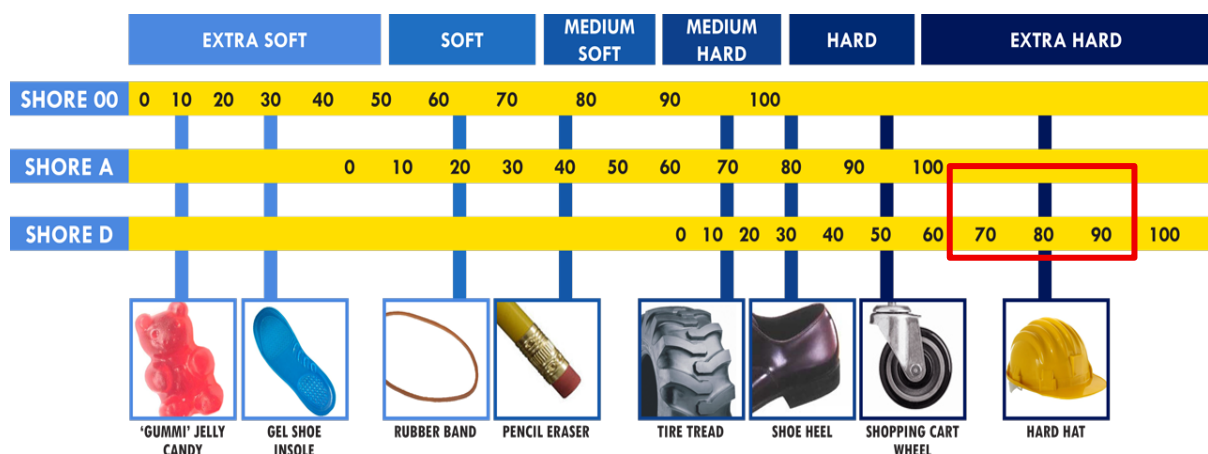
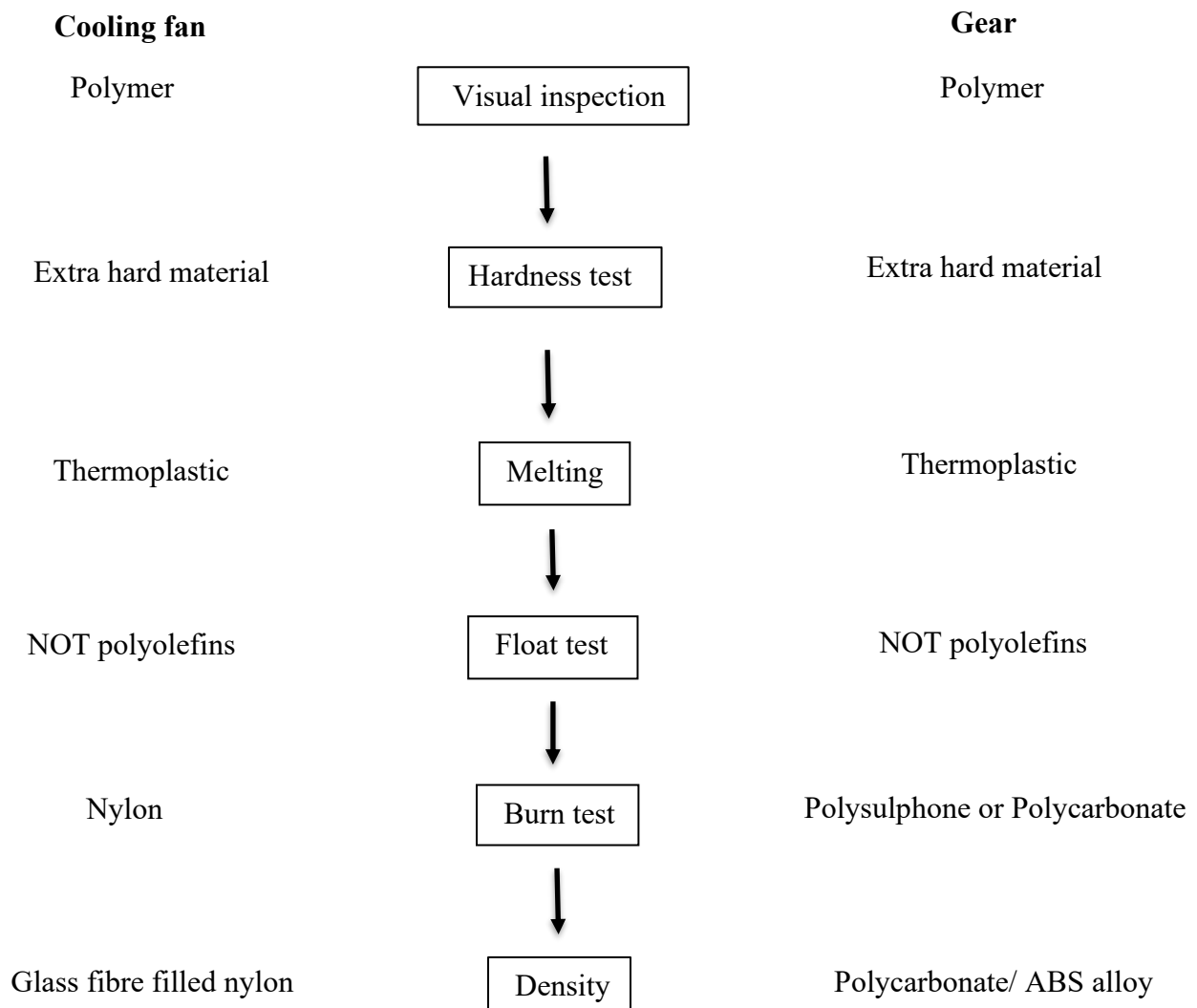


Figure 39 – *Hardness chart for plastics*

Both components have Shore D hardness > 70 – **extra hard**

PLASTICS IDENTIFICATION FLOW CHART

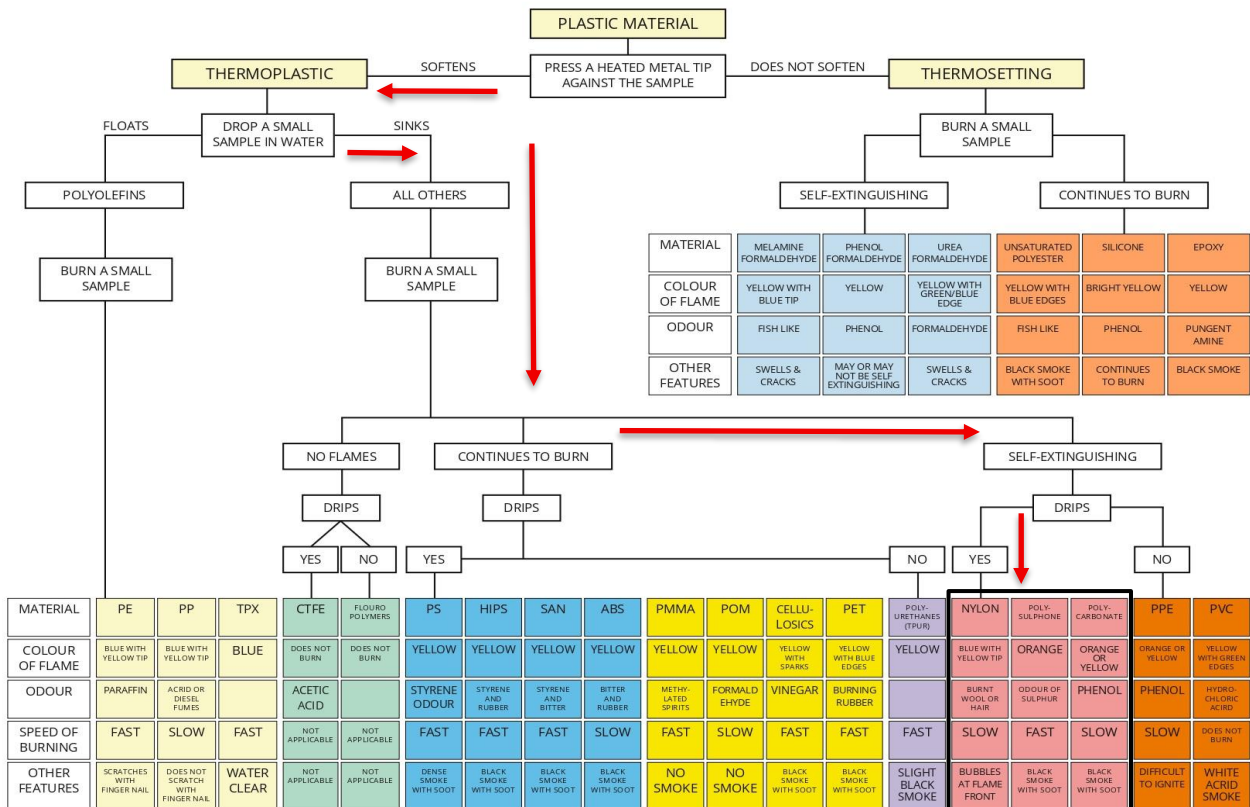


Figure 40 – Plastic identification chart for polymers

NYLON	POLY-SULPHONE	POLY-CARBONATE
BLUE WITH YELLOW TIP	ORANGE	ORANGE OR YELLOW
BURNT WOOL OR HAIR	ODOUR OF SULPHUR	PHENOL
SLOW	FAST	SLOW
BUBBLES AT FLAME FRONT	BLACK SMOKE WITH SOOT	BLACK SMOKE WITH SOOT

Cooling fan - polysulphone or polycarbonate

Gear - nylon

Overview of materials for Polycarbonate/ABS Alloy, Unreinforced

Categories: [Polymer](#); [Thermoplastic](#); [ABS Polymer](#); [Polycarbonate/ABS Alloy, Unreinforced](#); [Polycarbonate \(PC\)](#)

Material Notes:

This property data is a summary of similar materials in the MatWeb database for the category "Polycarbonate/ABS Alloy, Unreinforced". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors:

[Click here](#) to view all available suppliers for this material.

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	0.960 - 1.50 g/cc	0.0347 - 0.0542 lb/in ³	Average value: 1.16 g/cc Grade Count:1105
	0.935 - 1.01 g/cc @Temperature 240 - 260 °C	0.0338 - 0.0365 lb/in ³ @Temperature 464 - 500 °F	Average value: 0.981 g/cc Grade Count:10

Figure 41 – [MatWeb data for Polycarbonate/ ABS Alloy](#)

Cooling fan - Polycarbonate/ ABS

Overview of materials for Nylon 6, 30% Glass Fiber Filled

Categories: [Polymer](#); [Thermoplastic](#); [Nylon \(Polyamide PA\)](#); [Nylon 6 \(PA6\)](#); [Nylon 6, 30% Glass Fiber Filled](#)

Material Notes:

This property data is a summary of similar materials in the MatWeb database for the category "Nylon 6, 30% Glass Fiber Filled". Specific grades with glass content between 25% and 34% are included. Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors:

[Click here](#) to view all available suppliers for this material.

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	1.20 - 1.67 g/cc	0.0434 - 0.0603 lb/in ³	Average value: 1.37 g/cc Grade Count:1119
	1.136 - 1.20 g/cc @Temperature 220 - 270 °C	0.04104 - 0.0434 lb/in ³ @Temperature 428 - 518 °F	Average value: 1.17 g/cc Grade Count:10

Figure 42 – [Matweb data for Glass fibre filled Nylon 6](#)

Gear - Glass fibre filled Nylon

2.3 Improved material choice for gear manufacturing

POM (polyoxymethylene/ Acetal/ Delrin)

Pros

- Lower moisture absorption (~0.2%), high dimensional stability
- Stable performance across wide temperature range
- High stiffness
- Low friction, good self-lubrication
- Good chemical resistance
- High wear resistance
- Excellent machinability

Cons

- Lower impact strength than nylon
- More expensive than nylon
- Limited performance at very high temperatures

2.4 3D model and 2D drawings

- Measuring instruments – Vernier calliper, Micrometre screw gauge, Measuring pin set

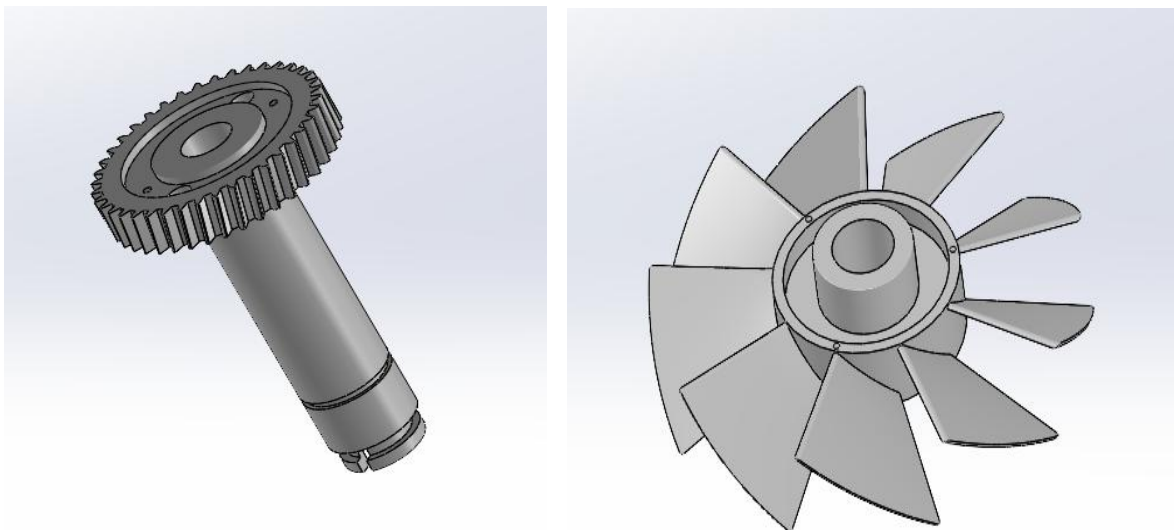


Figure 43 – 3D model of the parts

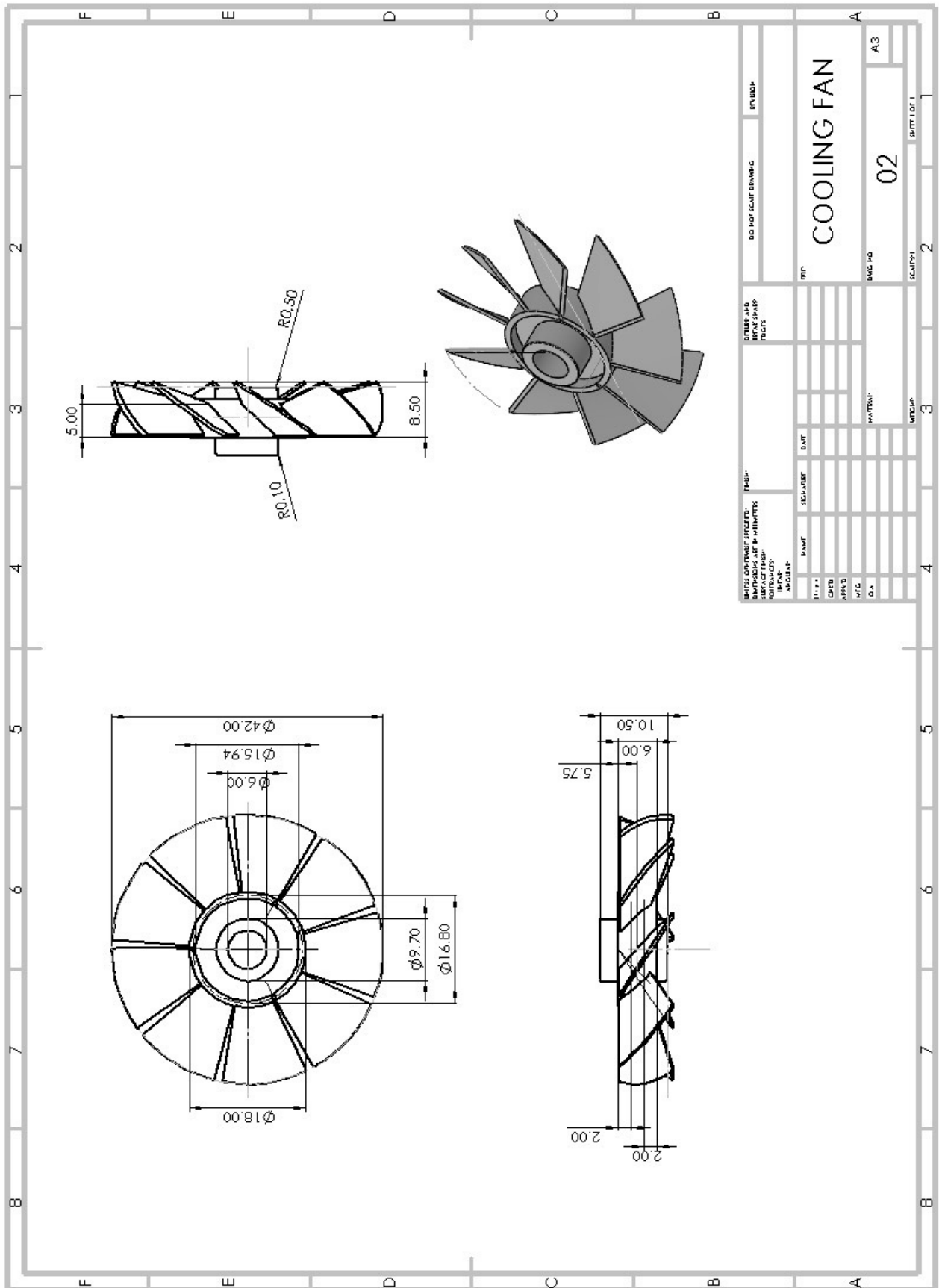


Figure 45 – SolidWorks of the cooling fan

2.5 Manufacturing methods

2.5.1 Possible Manufacturing methods

Gear manufacturing

- Injection moulding
- CNC moulding

cooling fan

- Injection moulding

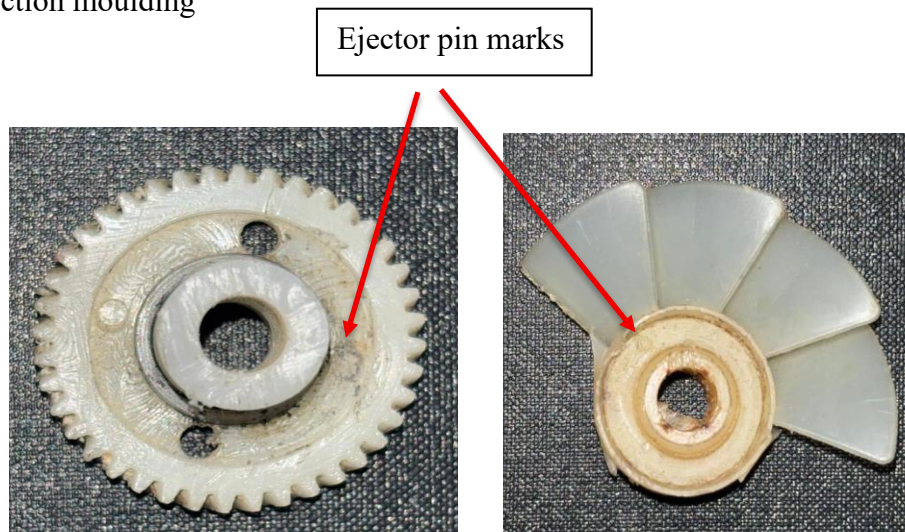


Figure 46 – Gear and cooling fan with ejector pin marks

Ejector pin marks observed on both components confirm manufacture by injection moulding

2.5.2 Why injection moulding?

- Produces complex shapes with high dimensional precision
- Consistent, identical parts suited to mass production
- Cost-effective for large-scale production
- Good surface finish with minimal post-processing required
- Minimal material waste

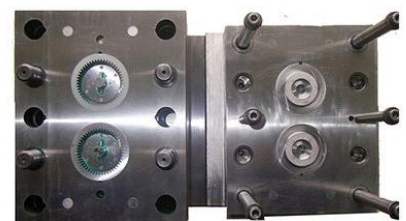


Figure 47 – Injection moulding of gears

2.5.3 Why CNC is not a suitable for gear manufacturing?

- slower production rates
- higher manufacturing costs
- More material waste

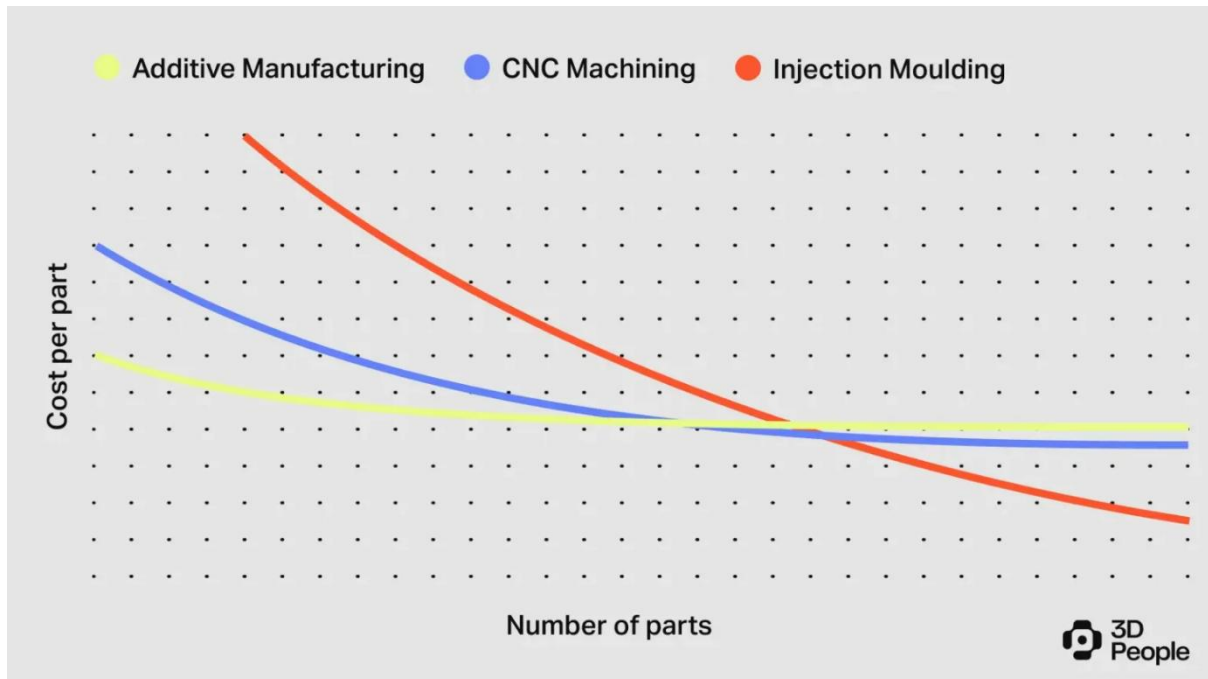


Figure 48 – *Graph of Cost per part Vs Number of parts for three types of plastic manufacturing methods*

Other manufacturing processes

Compression moulding	Poor dimensional accuracy for gears and thin blades
Extrusion	Only for continuous profiles, not for complex 3D design
Blow moulding	Mainly for hollow products
3D printing	Slower and weaker mechanical properties lower surface finish

2.5.4 Injection moulding

For mould design

- Aluminium moulds are used for prototyping and low-volume production.
- Tool steel moulds are used for high-volume production.
- Wire EDM is commonly used to machine hardened steel moulds for precise and complex gear profiles.

Material	Melt temperature °C	Mould temperature °C
Nylon	280 - 300	40 - 90
Polycarbonate	280 - 320	85 - 120

Effects of Injection Moulding on Polycarbonate and Nylon

Polycarbonate	Nylon
- High viscosity requires high moulding temperatures. - Moisture absorption can reduce surface and mechanical quality, if not dried properly. - Retains toughness, transparency, and dimensional stability.	- Requires controlled moulding due to shrinkage and filling issues. - Molecular and fibre alignment increases strength and stiffness. - Cooling crystallization improves hardness and wear resistance.

Selection of 2- or 3- plate moulds

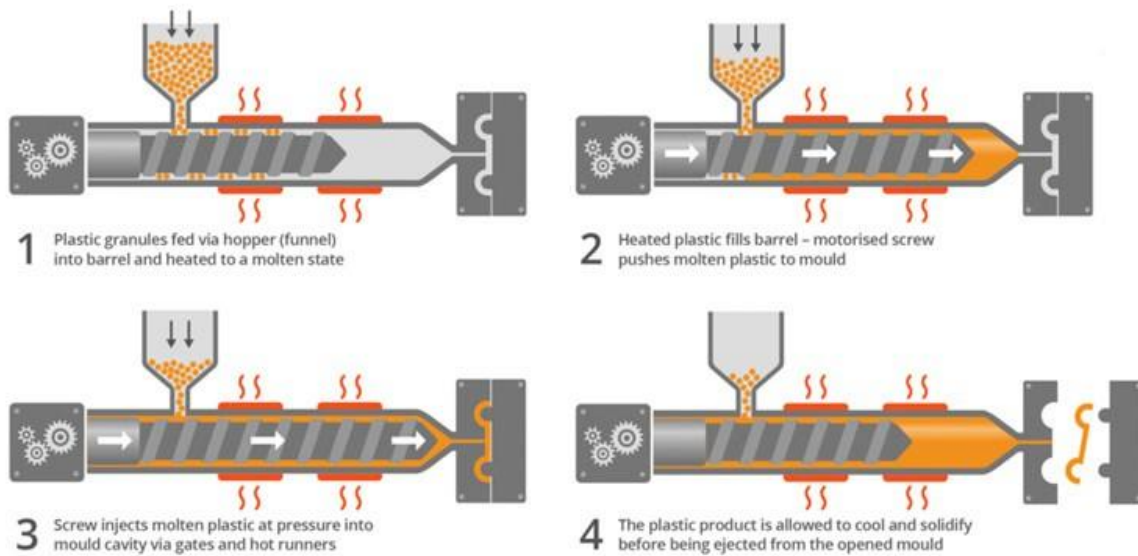


Figure 51 – Steps of injection moulding process

Parts are manufactured using 2- or 3-plate moulds depending on gating and precision needs. 3-plate moulds improve gate separation and appearance while 2-plate moulds are simpler, cheaper and more common.

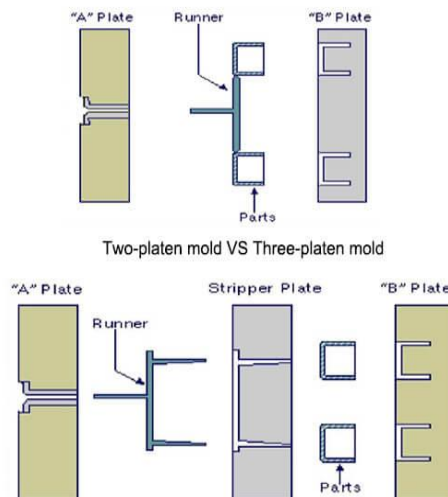


Figure 52 – parts of 2- and 3- plate injection moulding

2.5.5 Surface finish

Consistent machining marks observed ($R_a = 5.42$), likely from mould machining; left unpolished on non-functional areas to reduce cost.



Figure 53 – Machining like marks on the shaft of gear

Mitutoyo	Surtronic 3+
DATE	01-01-2000
TIME	00:25:44
STAND	JIS2001
PROFILE	R
FILTER	GAUSS
EVA-L	4.0mm
N	5
λ_c	0.8mm
λ_s	2.5mm
TILT-COMP.	ALL
M-SPEED	0.5mm/s
RANGE	AUTO
PRE/POST	ON
DRIVE	STAND
R-PROFILE	
EVA-L	4.0mm
λ_c	0.8mmX5
R_a	5.42 μm
R_z	21.87 μm
R_q	6.15 μm
R-PROFILE	
EVA-L	4.0mm
λ_c	0.8mmX5
	$\times 1K$
	$\times 50$
Ver.	10.0 $\mu\text{m}/\text{cm}$
Hor.	200.0 $\mu\text{m}/\text{cm}$

Figure 50 – Surface roughness data from Surtronic tester

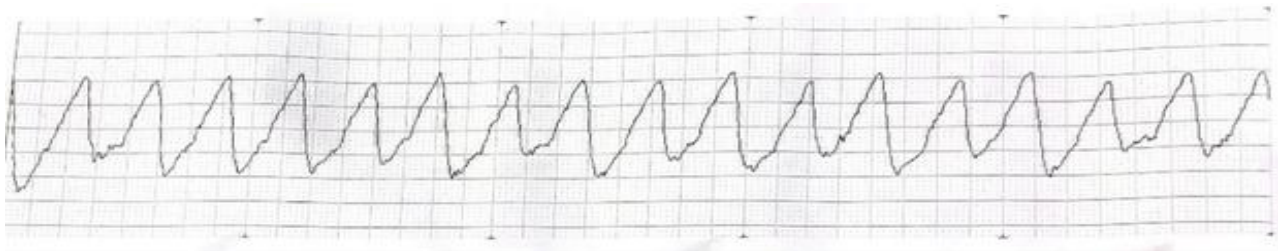


Figure 55 – Roughness pattern

Cooling fan has a smooth semi-gloss finish achieved using polished mould surfaces, improving appearance and part ejection. **SPI B-grade** polishing or **VDI 3400 textures (VDI 12–18)** are commonly used.



Figure 56 – SPI surface finish standard



Figure 57 – VDI surface finish standard

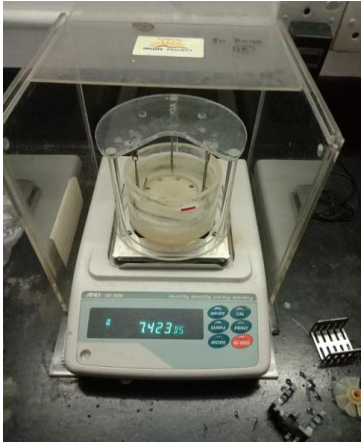
3 Beater of the hand mixer

3.1 Functions of the Beater

- Mixing and Blending: Uniformly combines wet and dry ingredients together into a single, consistent mixture.
- Aeration (Whipping): Incorporates air into the mixture as it rotates to increase volume and create foams (like whipped cream or egg whites).

3.2 Material identification process

Test	Observations and Conclusions
Visual Test  <i>Figure 58</i>	• The Beater material has Silver / Metallic colour and also has Highly polished, shiny, and reflective surface finish • So, the material can be stainless steel or chrome plated steel • Because there is no rust, the surface is actively protected by either the natural oxide layer of solid stainless steel or an undamaged chrome plating
Spark Test  <i>Figure 59</i>	• The material produced very few sparks when applied against the grinding wheel and the sparks that were visible appeared dull red or orange. • The sparks traveled in straight lines the sparks did not show any signs of branching or bursting. • The characteristics of the sparks indicate that the material has a very

	low carbon content and a high concentration of alloying elements.
Density Test  <i>Figure 60</i>	• The calculated density of the part was measured at 7.423 g/cm³. • This measured value is significantly higher than the density of lighter metals, such as aluminium or zinc. • The measured density is slightly lower than the theoretical ideal density of stainless steel (which is approximately 7.9 g/cm³). • The slight variation from the ideal density is attributed to standard experimental error, likely caused by tiny air bubbles clinging to the part or water displacement dynamics during the test. • The calculated density value definitively proves the part is made of steel and effectively rules out lighter metals.
Composition Test	• The chemical analysis shows the material is predominantly composed of iron (Fe), measured at an average of 80.9%. • There is a significant presence of chromium (Cr) alloyed into the material, measured at 14.20%. • The material possesses an exceptionally low carbon (C) concentration of 0.0287%.

- The material has a noticeably high concentration of sulfur (S), measured at 0.237%.
- The high iron and chromium content initially classifies the material as a stainless steel. Furthermore, the specific combination of very low carbon and intentionally high sulfur definitively identifies it as a free machining grade of stainless steel, which is specifically engineered to improve machinability during the manufacturing process.

Sample Result Name	Type	Method Name	Operator Name
Thymal -01/21	Unknown	Fe-01	
Measure Date Time	Instrument	Recalculation Date Time	
4/8/2026 6:56 PM	LMM01	4/8/2026 6:57 PM	
Sample Name	Counter	New	
Thymal -01	21		

			1	2	3	<X>	SD	RSD
C	%	Conc	0.0300	0.0297	0.0277	0.0287	0.0014	5.00
Si	%	Conc	0.244	0.164	0.128	0.146	0.0261	17.85
Mn	%	Conc	0.71	0.63	0.61	0.62	0.0173	2.78
P	%	Conc	0.0088	0.0120	0.0134	0.0127	0.0009	7.33
S	%	Conc	0.277	0.255	0.219	0.237	0.0259	10.93
Cr	%	Conc	14.37	14.15	14.26	14.20	0.082	0.58
Mo	%	Conc	0.255	0.255	0.256	0.256	0.0010	0.39
Ni	%	Conc	4.72	3.63	2.83	3.23	0.57	17.52
Al	%	Conc	[C<]0.0020	[C<]0.0020	[C<]0.0020	[C<]0.0020	0	0
Co	%	Conc	0.078	0.071	0.047	0.059	0.0175	29.73
Cu	%	Conc	0.125	0.121	0.121	0.121	0.0001	0.074
Nb	%	Conc	[C<]0.0050	[C<]0.0050	[C<]0.0050	[C<]0.0050	0	0
Ti	%	Conc	0.0233	0.0202	0.0177	0.0189	0.0017	9.07
V	%	Conc	0.110	0.101	0.100	0.101	0.0004	0.36
W	%	Conc	[C<]0.050	[C<]0.050	[C<]0.050	[C<]0.050	0	0
Pb	%	Conc	0.0177	0.0129	[C<]0.0100	0.0114	0.0020	17.90
Mg	%	Conc	0.0300	0.0269	0.0234	0.0252	0.0025	9.89
B	%	Conc	[C<]0.0010	[C<]0.0010	[C<]0.0010	[C<]0.0010	0	0
Fe	%	Conc	78.9	80.5	81.3	80.9	0.60	0.74

Figure 61 – Composition test results

3.3 Manufacturing methods

Possible manufacturing method for the beater part is CNC wire forming.

3.3.1 Process of CNC wire forming of a beater

CNC wire forming represents the industry-standard methodology for the high-volume manufacturing of traditional stainless steel wire beaters, offering excellent precision, scalability, and repeatability.

- **Wire Straightening:** Long, continuous coils of raw, food-grade stainless steel wire (typically 304 or 430 grade) are unspooled and passed through a series of mechanical straighteners to remove any inherent curvature and residual stresses from the coiling process.
- **Stamping/Cutting:** A progressive die machine precisely cuts the straightened wire to predetermined lengths. During this stage, it also flattens specific localized sections of the wire such as the central mating joints or the lower paddle areas to increase the effective surface area for both mixing performance and subsequent welding.
- **CNC Wire Forming:** The cut wires are fed into automated Computer Numerical Control (CNC) bending machines. Programmed robotic arms precisely bend the linear wire into the complex, multi-axial signature loop geometry required for optimal fluid agitation and aeration.
- **Resistance Welding:** Finally, the formed wire loops are permanently joined to the central drive shaft utilizing spot or resistance welding. This localized fusion technique creates a high-strength structural bond through the application of focused electrical current and pressure, ensuring a clean, hygienic connection without the introduction of secondary filler materials like solder.

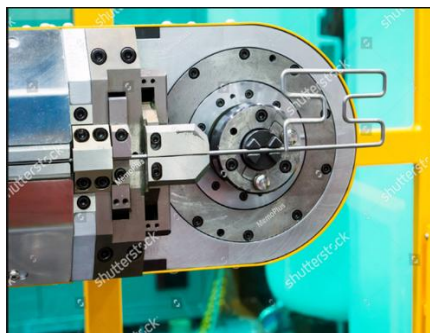


Figure 62

3.3.2 Finishing Processes

Regardless of the primary forming methodology, the manufactured beaters must undergo stringent surface treatment protocols to ensure food-grade compliance, aesthetic quality, and user safety.

1. Electropolishing: Electro polishing is an advanced electrochemical surface modification technique, often conceptualized as "reverse electroplating."

- **The Process:** The stainless-steel beater is submerged in a temperature-controlled, acidic electrolytic bath and connected as the anode. When a direct electrical current is applied, a microscopic, precise layer of the surface material is uniformly dissolved into the electrolyte.
- **The Purpose:** This process preferentially targets and flattens microscopic surface peaks, leaving the beater with an exceptionally smooth, ultra-reflective finish. Crucially for culinary applications, it eliminates micro-crevices and tiny pits, preventing the accumulation of bacteria and pathogens while significantly enhancing the material's passive resistance to corrosion.

2. Vibratory Finishing: Vibratory finishing (or mass finishing) is a mechanical surface treatment utilized primarily for deburring and edge radiusing.

- **The Process:** The raw, welded beaters are loaded into a large, high-frequency vibrating tub or bowl filled with specialized abrasive media (typically geometrically shaped ceramic or high-density plastic).
- **The Purpose:** The kinetic energy from the tub causes the media to continuously scrub and "tumble" against the beaters in a fluid-like motion. This abrasive action safely and effectively grinds away any sharp, jagged burrs, slag, or sharp edges left over from the progressive die cutting and resistance welding stages, ensuring the final product is safe to handle and structurally uniform.

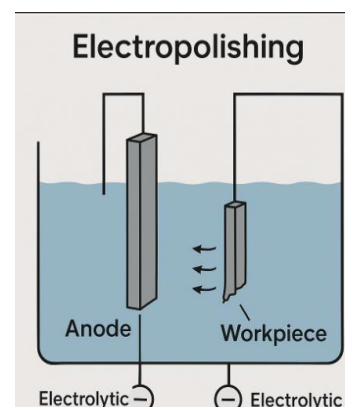


Figure 63

3.4 2D Drawing and 3D Modelling of the beater

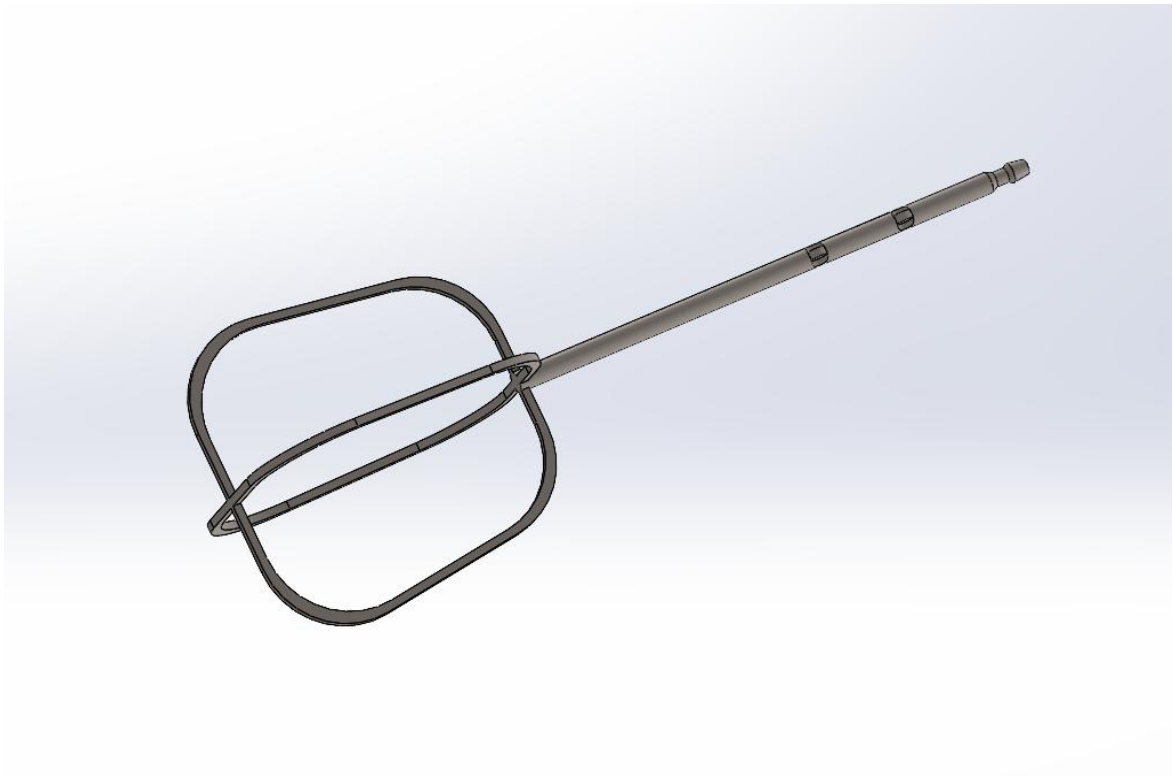


Figure 64

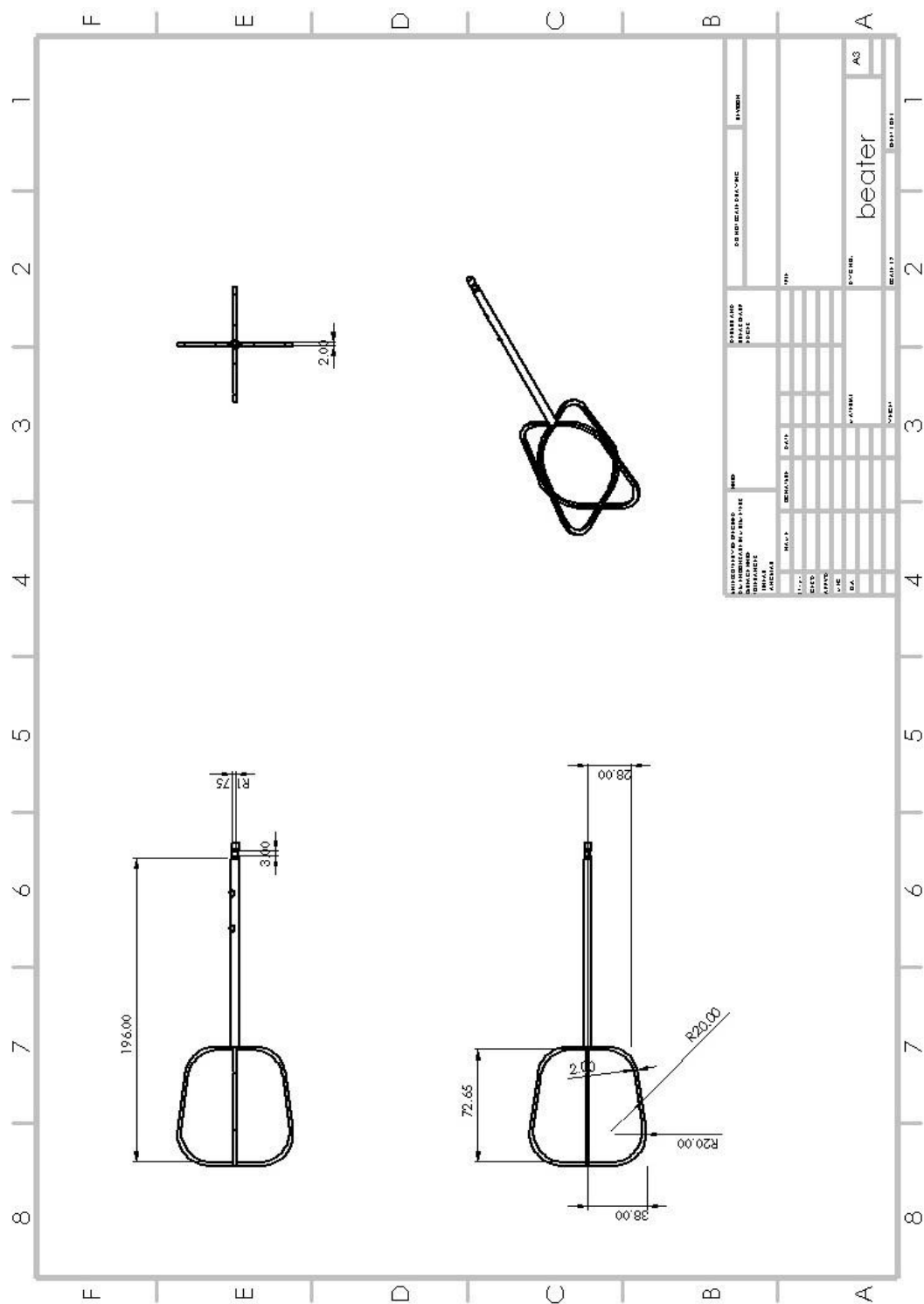


Figure 65

3.5 Improvements suggested for the beater

1. Advanced Material Composition Transitioning the primary construction material from traditional plated steel to advanced engineering polymers, specifically utilizing reinforced nylon and high-strength composites.

- **Suggested modification:** Replacing standard metallic wire forms with structurally optimized, injection-molded composite materials.
- **The Engineering Benefit:** This material upgrade provides a superior strength-to-weight ratio, which significantly reduces the rotational inertia of the beater assembly and lowers the continuous torque demand on the mixer's motor. Furthermore, solid composites eliminate the risk of surface plating delamination (flaking) into the mixture after prolonged wear, ensuring better long-term food safety while maintaining excellent fatigue resistance under cyclic loading.

2. Optimized Surface Area via Helical Geometry Redesigning the standard overlapping wire loops into a continuous helix (spiral) structure to maximize mechanical engagement with the fluid.

- **Suggested modification:** Implementing a helical structural frame that substantially increases the active "contact zone" (swept volume) of the beater as it rotates through a viscous medium.
- **The Engineering Benefit:** The complex geometry of the helix induces a highly efficient, multi-directional flow pattern (simultaneous axial and radial fluid displacement). This continuous, sweeping overlap minimizes stagnation or "dead zones" within the mixing bowl, resulting in faster homogenization, more consistent blending, and superior aeration compared to traditional planar wire designs.



Figure 66

4 Motor Housing

4.1 Functional properties of the motor housing in hand mixer.

Motor housing is a critical structural component of a hand mixer. It serves as the primary enclosure for the electric motor and performs several essential functions that ensure reliable performance, durability, and safety during operation.

1. Withstand Motor Loads
2. Vibration Resistance
3. Maintain Motor Alignment
4. Heat Dissipation

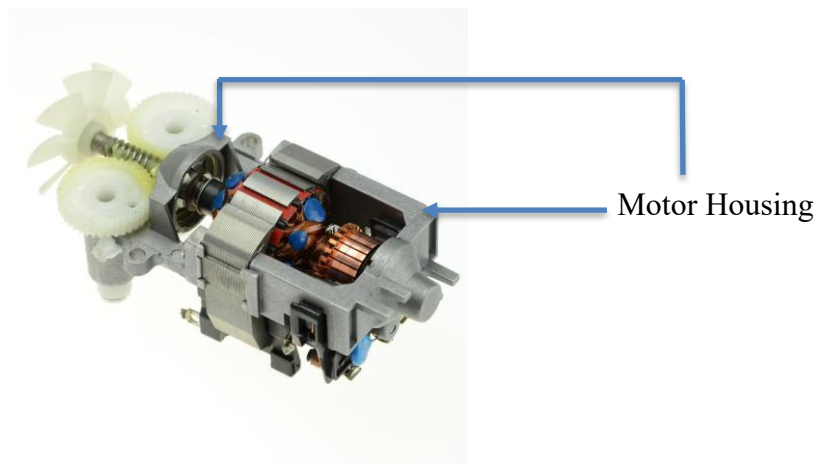


Figure 67 - Motor and gear assembly with motor housing

4.2 Product Analysis of Motor Housing.

For material identification used in motor housing three tests were conducted.

Density test

Objective

The density test was performed to determine the density of the material used in the motor housing of the hand mixer.

Procedure

The density was measured using a digital density meter, which provides the specific gravity (relative density) of the material. The actual density value was then obtained by multiplying the specific gravity by the standard density of water (1.0 g/cm^3) at room temperature.

Result

Density of the material is obtained as 5.055 g/cm^3 .

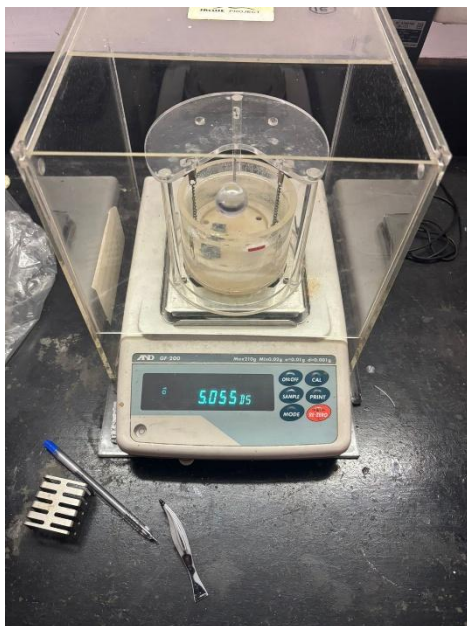


Figure 68 - Density meter measurement

Conclusions

Typical Material Properties: Zn and ZA Alloys					cut from production die castings.		
Commercial:	Zamak Die Casting Alloys				ZA Die Casting Alloys		
	No. 2	No. 3 AG-40A	No. 5 AG-41A	No. 7 AG-40B	ZA-8	ZA-12	ZA-27
Mechanical Properties							
Ultimate Tensile Strength							
As-Cast ksi (MPa)	52 (359)	41 (283)	48 (331)	41 (283)	54 (374)	59 (404)	62 (426)
Aged ksi (MPa)	48 (331)	35 (241)	39 (269)	41 (283)	43 (297)	45 (310)	52 (359)
Yield Strength ^(A)							
As-Cast ksi (MPa)	41 (283)	32 (221)	33 (228)	32 (221)	42 (290)	46 (320)	54 (371)
Aged ksi (MPa)					32 (224)	35 (245)	46 (322)
Compressive Yield Strength ^(B)							
As-Cast ksi (MPa)	93 (641)	60 (414) ^(C)	87 (600) ^(C)	60 (414) ^(C)	37 (252)	39 (269)	52 (358)
Aged ksi (MPa)	93 (641)	60 (414)	87 (600)	60 (414)	25 (172)	27 (186)	37 (255)
Elongation							
As-Cast % in 2 in. (51mm)	7	10	7	13	8	5	2.5
Aged % in 2 in. (51mm)	2	16	13	18	20	10	3
Hardness ^(D)							
As-Cast BHN	100	82	91	80	103	100	119
Aged BHN	98	72	80	67	91	91	100
Shear Strength							
As-Cast ksi (MPa)	46 (317)	31 (214)	38 (262)	31 (214)	40 (275)	43 (296)	47 (325)
Aged ksi (MPa)	46 (317)	31 (214)	38 (262)	31 (214)	33 (228)	33 (228)	37 (255)
Impact Strength							
As-Cast ft-lb (J)	35 (47.5)	43 ^(E) (58)	48 ^(E) (65)	43 ^(E) (58)	31 ^(E) (42)	21 ^(E) (29)	9 ^(E) (13)
Aged ft-lb (J)	5 (7)	41 (56)	40 (54)	41 (56)	13 (18)	14 (19)	3.5 (5)
Fatigue Strength ^(F)							
As-Cast ksi (MPa)	8.5 (58.6)	6.9 (47.6)	8.2 (56.5)	6.8 (46.9)	15 (103)	17 (117)	21 (145)
Aged ksi (MPa)	8.5 (58.6)	6.9 (47.6)	8.2 (56.5)	6.8 (46.9)	15 (103)	—	21 (145)
Young's Modulus							
psi x 10 ⁶ (GPa)	^(G)	^(G)	^(G)	^(G)	12.4 (85.5)	12 (82.7)	11.3 (77.9)
Physical Properties							
Density							
lb/in ³ (g/cm ³)	0.24 (6.6)	0.24 (6.6)	0.24 (6.6)	0.24 (6.6)	0.227 (6.3)	0.218 (6.03)	0.181 (5.000)
Melting Range							
°F (°C)	715-734 (379-390)	718-728 (381-387)	717-727 (380-386)	718-728 (381-387)	707-759 (375-404)	710-810 (377-432)	708-903 (375-487)
Specific Heat							
BTU/lb °F (J/kg °C)	0.10 (419)	0.10 (419)	0.10 (419)	0.10 (419)	0.104 (435)	0.107 (450)	0.125 (525)
Coefficient of Thermal Expansion							
μ in/in°F (μ m/m°K)	15.4 (27.8)	15.2 (27.4)	15.2 (27.4)	15.2 (27.4)	12.9 (23.2)	13.4 (24.2)	14.4 (26.0)
Thermal Conductivity							
BTU/ft hr°F (W/m °K)	60.5 (104.7)	65.3 (113)	62.9 (109)	65.3 (113)	66.3 (114.7)	67.1 (116.1)	72.5 (122.5)
Electrical Conductivity							
μ Ω in.	25.0	27.0	26.0	27.0	27.7	28.3	29.7
Poisson's Ratio							
	0.30	0.30	0.30	0.30	0.30	0.30	0.30

Figure 69 - density comparison of zinc alloys

According to the data the result matches the density range of Zinc Aluminum alloys. **Most** closets material can be ZA27.

Spark Test

Objective

Categories whether material is ferrous or nonferrous.

Procedure

A portion of the motor housing was ground using an appropriate grinding tool, and the resulting spark pattern (or absence of sparks) was carefully observed.

Result

No sparks were generated during the grinding process.



Figure 70 - Spark Test

Hence the material should be non-ferrous material.

Spark-Optical Emission Spectrometry (OES)

Objective

Identify the alloys composition.

Procedure

Specimen was tested under OES tester which generated the results of alloy composition by observing the light spectrum when subjected to chemical burning.

Result

Obtained results were not clear. An analytic error was detected by the computer.

Wednesday, April 8, 2026 7:13:31 PM

Sample Result Name	Type	Method Name	Operator Name
Dasun Sanjaya/28	Unknown	Fe-01	

Measure Date Time	Instrument	Recalculation Date Time
4/8/2026 7:12 PM	LMM01	4/8/2026 7:13 PM

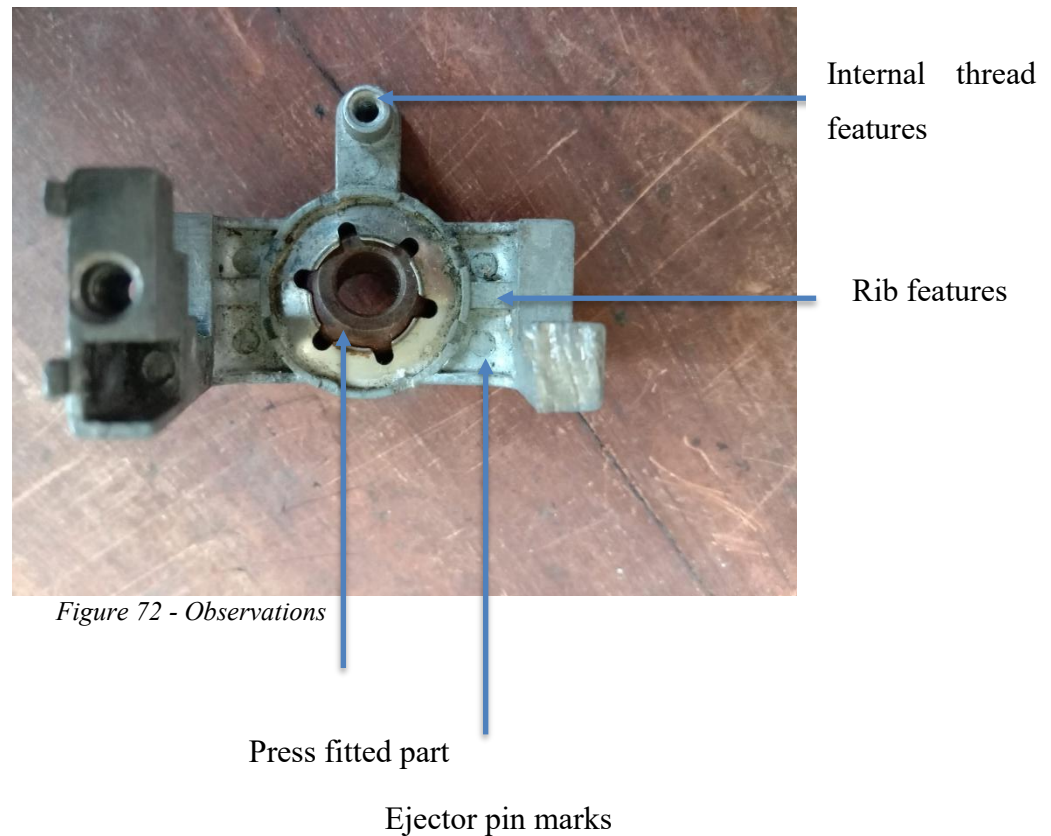
Sample Name	Counter	New
Dasun Sanjaya	28	

		1	2	<X>	SD	RSD
C	% Conc	[Analyt!] 0.41	[Analyt!] 0.158	[Analyt!] 0.284	0.178	62.7
Si	% Conc	[Analyt!] 0.058	[Analyt!] 0.048	[Analyt!] 0.053	0.0068	12.77
Mn	% Conc	[Analyt!] 0.0188	[Analyt!] 0.0166	[Analyt!] 0.0177	0.0016	8.86
P	% Conc	[Analyt!] 0.0063	[Analyt!] 0.0055	[Analyt!] 0.0059	0.0006	10.40
S	% Conc	[Analyt!] 0.0067	[Analyt!] 0.0055	[Analyt!] 0.0061	0.0008	13.25
Cr	% Conc	[Analyt!] 0.0192	[Analyt!] 0.0161	[Analyt!] 0.0177	0.0022	12.44
Mo	% Conc	[Analyt!] 0.0282	[Analyt!] 0.046	[Analyt!] 0.037	0.0124	33.58
Ni	% Conc	[Analyt!] 0.092	[Analyt!] 0.094	[Analyt!] 0.093	0.0009	0.93
Al	% Conc	[C>] [Analyt!] 2.85	[C>] [Analyt!] 2.85	[C>] [Analyt!] 2.85	0	0
Co	% Conc	[Analyt!] 0.0271	[Analyt!] 0.0228	[Analyt!] 0.0249	0.0030	12.04
Cu	% Conc	[Analyt!] 0.0139	[Analyt!] 0.0127	[Analyt!] 0.0133	0.0008	6.14
Nb	% Conc	[Analyt!] 0.036	[Analyt!] 0.0291	[Analyt!] 0.032	0.0047	14.58
Ti	% Conc	[Analyt!] 0.0271	[Analyt!] 0.0222	[Analyt!] 0.0247	0.0035	14.09
V	% Conc	[Analyt!] 0.0170	[Analyt!] 0.0138	[Analyt!] 0.0154	0.0022	14.61
W	% Conc	[Analyt!] 0.38	[Analyt!] 0.30	[Analyt!] 0.34	0.054	16.02
Pb	% Conc	[C<] [Analyt!] 0.0100	[C<] [Analyt!] 0.0100	[C<] [Analyt!] 0.0100	0	0
Mg	% Conc	[Analyt!] 0.167	[Analyt!] 0.144	[Analyt!] 0.156	0.0160	10.29
B	% Conc	[C<] 0.0010	[C<] 0.0010	[C<] 0.0010	0	0
Fe	% Conc	95.8	96.2	96.0	0.222	0.232

Figure 71 - Test results obtained at Spark OES test

4.3 Manufacturing Methods

Based on the following observations there are three manufacturing methods proposed associated with motor hosing.

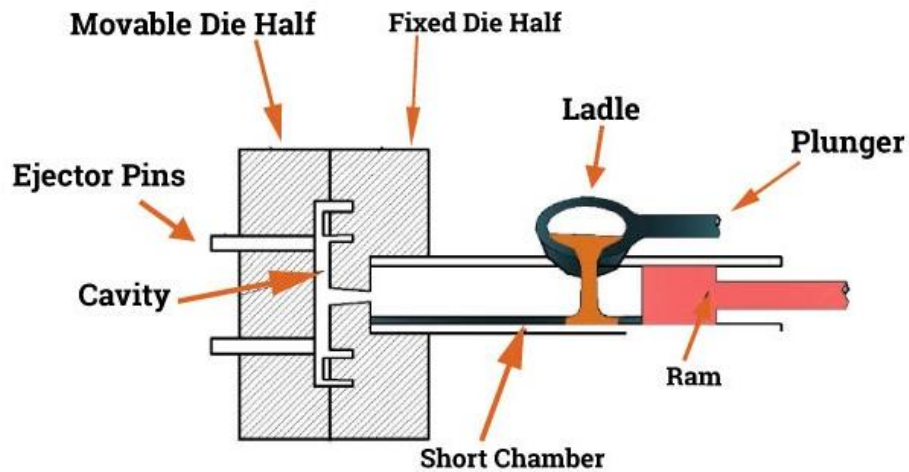


1. Die casting

Due to the continuous and complex geometry of motor housing, the most suitable manufacturing method is casting. The presence of clear ejector pin marks on the component indicates that the process used was die casting. Additionally, the incorporation of rib features and filleted edges enhances the overall quality of the cast part.

Material Compatibility

The identified material belongs to the non-ferrous category. Since ZA-27 has a relatively high melting point, the appropriate die casting process is Cold Chamber Die Casting.



Schematic of cold chamber die casting process

COMPARISON RATINGS	ZA-8	ZA-12	ZA-27
SAND CASTABILITY	G	E	F
PERM. MOLDABILITY	E	E	F
DIE CASTABILITY	E	G	G
STRENGTH	G	G	E
DUCTILITY	G	G	G
BEARING WEAR	G	E	E
MACHINABILITY	E	E	E
PRESSURE TIGHTNESS	G	E	F
PLATING	E	G	F
ZINC ANODIZING	E	E	E
CHROMATING	E	G	F
PAINTING	E	E	E
DIMENSIONAL STABILITY	G	E	F
E-Excellent G-Good F-Fair			

Figure 73 - Comparison between zinc aluminum alloys' properties

2.Tapping

To achieve higher production rate high speed tapping may be used as the manufacturing of the internal thread features.



Figure 74 - High speed tapping machine

3.Press fitting

For the ring inside the motor housing which is used to align the motor shat is embedded into the housing using press fitting.

4.Finishing methods.

Polishing operation was conducted to get the surface finish of the component. This can be categories into after manufacturing method.



Figure 75 - Touch comparison test to identify the finishing operation.

Casting Process	Comment
Sand Casting	Surface finishing is not good. cannot achieve high production rate. Getting dimensional accuracy will be costly.
Hot chamber die casting	Since the melting point of used material is higher hot chamber die casting cannot be done.
Investment Casting	The process is not economical and cannot achieve the required production rate.

Table 3-16 Competitive Performance Chart

Alloy Property	ZA- MAK 3**	ZA- MAK 5**	ZA-8**			ZA-12**			ZA-27**	
	Die Cast	Die Cast	Sand Cast	Perm Mold	Die Cast	Sand Cast	Perm Mold	Die Cast	Sand Cast	Perm Mold
Mechanical Properties										
Ultimate Tensile Strength										
psi x10 ³ (MPa)	41 (283)	48 (331)	38 (263)	35 (240)	54 (374)	43 (299)	48 (328)	59 (404)	61 (421)	64 (441)
Yield Strength										
psi x10 ³ (MPa)	32 (221)	33 (228)	29 (198)	30 (208)	42 (290)	31 (211)	39 (268)	46 (320)	54 (371)	55 (376)
Elongation										
% in 2in.	10	7	1.7	1.3	8	1.5	2.2	5	4.6	2.5
Young's Modulus										
psi x10 ⁶ (MPa x 10 ³)	12.4*** (85.5)	12.4*** (85.5)	12.4 (85.5)	12.4 (85.5)	12.4 (85.5)	12.0 (82.7)	12.0 (82.7)	12.0 (82.7)	11.3 (77.9)	11.3 (77.9)
Torsional Modulus										
psi x10 ⁶ (MPa x 10 ³)	4.8 (33.1)	4.8 (33.1)	4.8 (33.1)	4.8 (33.1)	4.8 (33.1)	4.6 (31.7)	4.6 (31.7)	4.6 (31.7)	4.3 (29.6)	4.3 (29.6)
Shear Strength										
psi x10 ³ (MPa)	31 (214)	38 (262)	N/A	35 (241)	40 (275)	37 (253)	35 (241)	43 (296)	42 (292)	N/A
Hardness										
(Brinell)	82	91	85	87	103	94	89	100	113	114
Impact Strength										
ft-lb (J)	43 (58)	48 (65)	15 (20)	N/A	31 (42)	19 (25)	N/A	21 (29)	35 (48)	N/A
Fatigue Strength Rotoary Bedn (5 x 10⁶ cycles)										
psi x10 ³ (MPa)	6.9 (47.6)	8.2 (56.5)	N/A	7.5 (57.1)	15 (103)	15 (103)	N/A	17 (117)	25 (172)	N/A
Compressive Yield Strength 0.1% Offset										
psi x10 ³ (MPa)	60 (414)	87 (600)	29 (199)	31 (210)	37 (252)	33 (230)	34 (235)	39 (269)	48 (330)	N/A

* Minimum Properties

** Complies with ASTM specification B86.

*** Varies with stress level; applicable only for shot-duration loads.

Figure 76 - Material property change with different casting methods

ZA-27 Zinc-Aluminium Alloy shows improved strength and hardness when manufactured using permanent mould casting compared to sand casting. Faster cooling in permanent mould casting produces a finer microstructure, increasing tensile and yield strength. However, sand casting provides higher elongation, giving better ductility, while elastic properties remain nearly unchanged between the processes.

4.4 Measurement Process



Figure 77 – Component measurements

Measuring Instruments

1. Digital vernier caliper.
2. Digital micrometer screw gauge.
3. Height gauge.
4. Fillet gauges.

4.5 3D CAD model

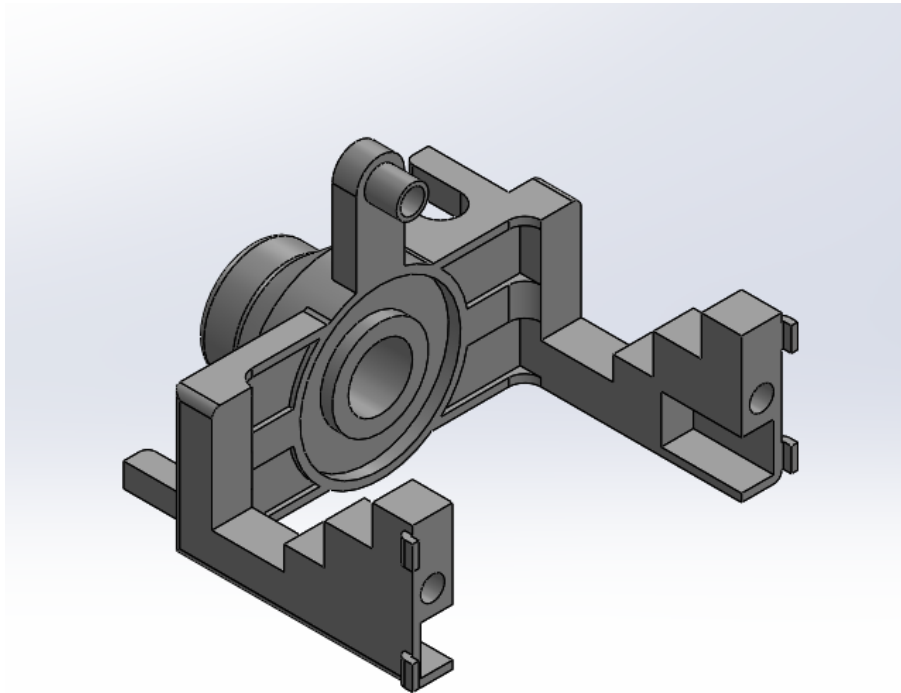


Figure 51 - Back part of the motor housing

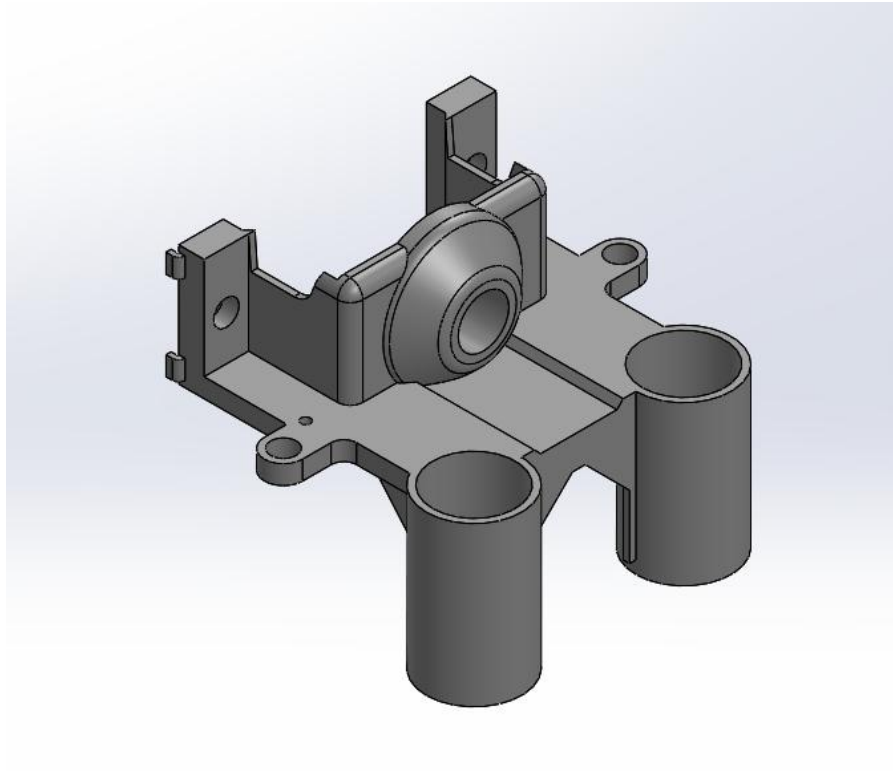
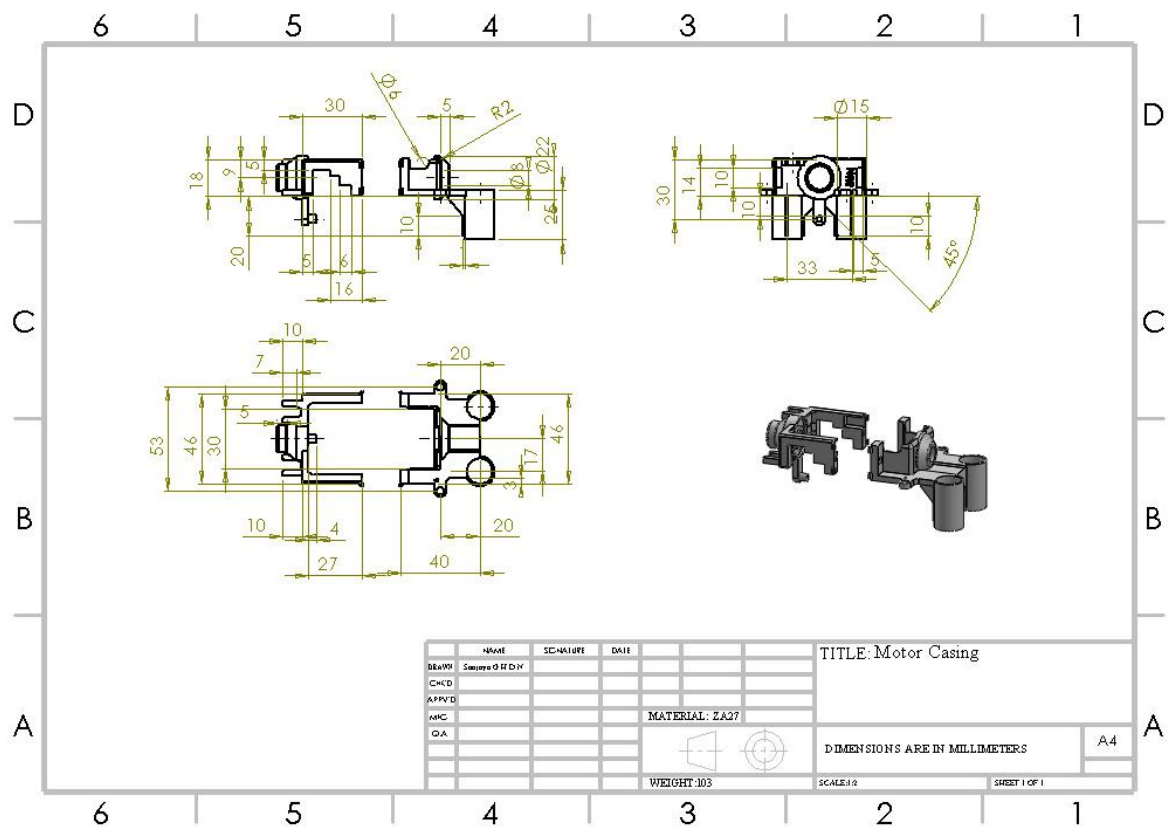


Figure 52 - Front part of the motor housing

4.6 Drawing



5 Hand Mixer Motor Shaft

5.1 Functions and features of the hand mixer



Figure 53 - – shaft with worm gear and press fit ball

The selected component for reverse engineering was the motor output shaft with a worm gear from a hand mixer. The shaft is directly connected to the motor armature and is responsible for transmitting rotational power from the motor to the gear mechanism.

Main functions of the shaft

- Transmit rotational motion from motor to gears
- Reduce speed and increase torque using worm gear mechanism
- Change power transmission direction by 90°
- Support smooth rotational movement
- Hold cooling fan and worm gear assembly
- Provide thrust support using press-fitted steel ball

The worm gear mechanism is used because it provides high torque at low speed in a compact space. The press-fitted steel ball at the end of the shaft acts as a thrust support and reduces

friction during operation. The component must therefore possess good torsional strength, wear resistance, dimensional accuracy, and smooth surface finish for reliable operation.

5.2 Material Identification

Material identification was carried out using density and spark testing methods.

Density Test

- Measured density $\approx 7.289 \text{ g/cm}^3$
- Density falls within steel material range

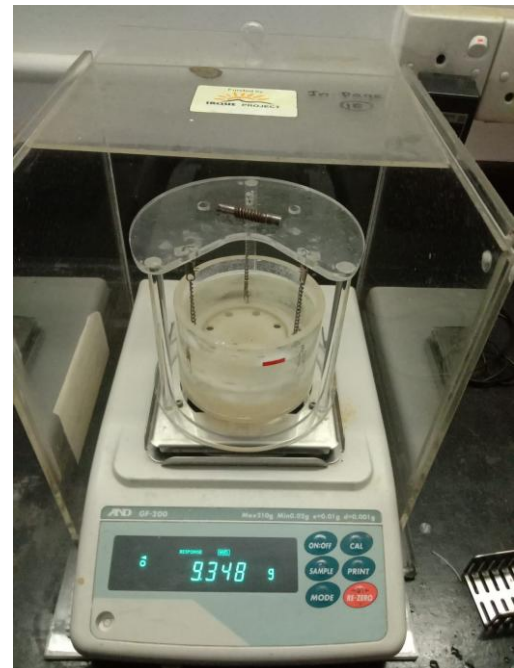


Figure 54 - density meter with the sample

Spark Test

Spark Test Observations

- Yellow-orange sparks observed
- Moderate spark branching detected

Indicates low carbon ferrous material

Based on the density result, spark characteristics, application requirements, and manufacturing suitability, the shaft material was identified as **mild steel (low carbon steel)** [1], [2].



Figure 55 – bench grinder

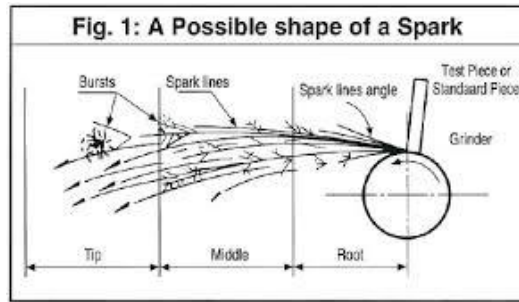


Figure 83 – spark shape

Metal Identification by Spark Test with either a portable or stationary grinder.

METAL	COLOR of Stream		Stream		Spurts	
	NEAR WHEEL	NEAR END	Volume	Length in/mm	Quantity of	Nature of
Wrought Iron	Straw	White	Large	65/1651	Very Few	Forked
1020 steel	White	White	Large	70/1778	Few	Forked
Carbon Tool Steel	White	White	(M)Large	55/1397	Very Many	Fine repeating
Gray Cast Iron	Red	Straw	Small	25/635	Many	Fine repeating
White Cast Iron	Red	Straw	Very small	20/508	Few	Fine repeating
Annealed Malleable cast	Red	Straw	Moderate (M)	30/762	Many	Fine repeating
High Speed steel	Red	Straw	Small	60/1524	Very Few	Forked
Manganese steel	White	White	(M)Large	45/1143	Many	Fine repeating
Stainless	Straw	White	Moderate (M)	50/1270	Many	Fine repeating
Tungsten Cr Die steel	Red	Straw	Small	35/889	Many	Fine repeating
Nitrided nitralloy	White	White	Large curved	55/1397	Moderate	Forked
Stellite	Orange	Orange	Very small	10/254	none	
Cemented tungsten carbide	Light Orange	Light Orange	Extra small	2/50.8	none	
Nickel	Orange	Orange	Very small	10/254	none	
Copper, brass, Aluminum			none		none	

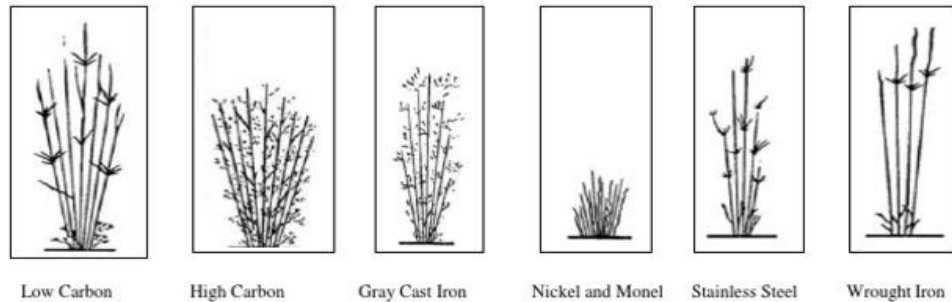


Figure 84 – metal identification spark test chart

Reasons for selecting mild steel

- Good ductility and toughness
- Adequate torsional strength for low-power application
- Easy to machine and grind
- Suitable for thread rolling and press fitting
- Low manufacturing cost
- Good mass production capability

Mild steel is preferred over fully hardened or alloy steels because the shaft requires a tough core to resist torsional shock during operation. Excessively hard materials may become brittle and crack under repeated loading conditions.

5.3 Identified Manufacturing Methods

Visual inspection and dimensional analysis were used to identify the manufacturing methods used for the shaft.

Identified manufacturing methods

- Turning
- Grinding
- Thread cutting / worm machining
- Press fitting

The cylindrical shaft surfaces appear smooth and concentric, indicating turning and grinding operations. The worm profile contains sharp thread edges, suggesting that the worm gear was produced using thread cutting or machining operations.

Ball fitting observations

- No welding marks observed
- Ball fitted using interference/press fit
- Provides compact assembly and low friction support

The existing manufacturing process is suitable for economical mass production of small appliance components [3].

5.4 3d model and drawing

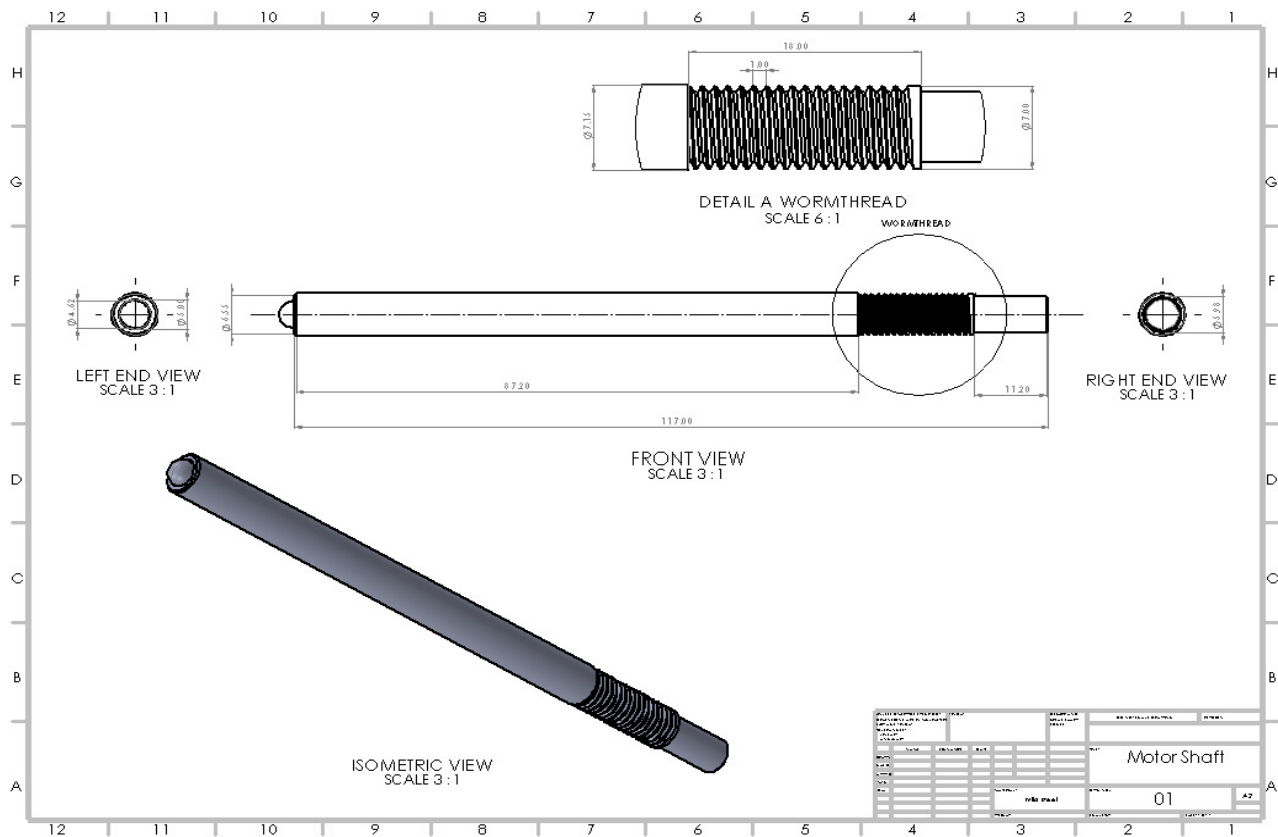


Figure 56 - 3d model and drawing

5.5 Proposed Improvements

(a) Thread Rolling Instead of Thread Cutting

The main proposed improvement is replacing thread cutting with thread rolling for the worm region.

Advantages of thread rolling

- No material removal
- Better surface finish
- Faster production rate
- Reduced material waste
- Improved dimensional consistency
- Better fatigue strength due to continuous grain flow

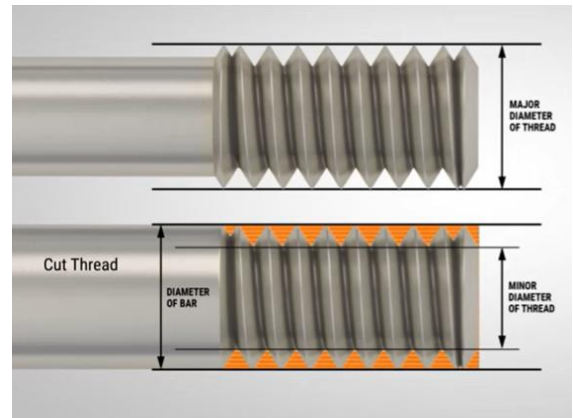


Figure 57 - thread material loss area

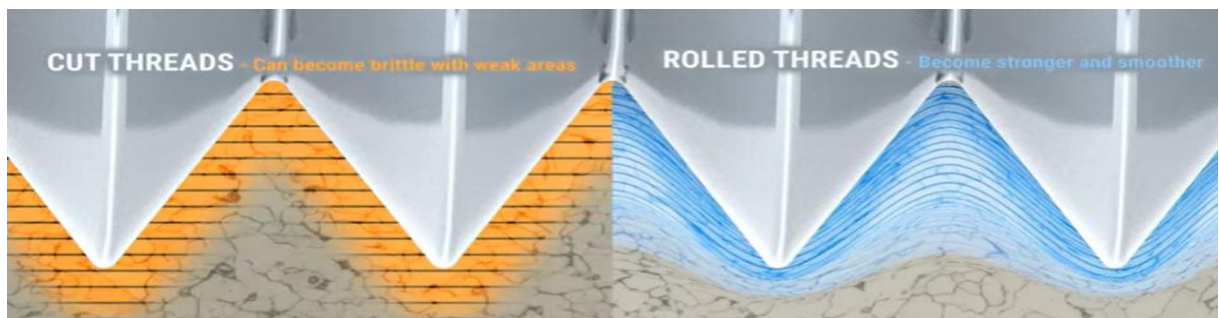


Figure 58 - difference of rolling and machining

Thread rolling produces compressive surface stresses, which improve wear resistance and torsional fatigue life [4]. The smoother worm surface also reduces friction and noise during gear meshing with the plastic driven gear.

(b) Surface Hardening of Worm Region

Another improvement is applying carburizing or case hardening only on the worm surface.

Benefits of carburizing

- Hard wear-resistant outer surface
- Tough and ductile mild steel core retained
- Improved gear life
- Better resistance to surface wear
- Reduced risk of brittle fracture

During carburizing, carbon diffuses into the outer surface layer. After quenching, the outer layer transforms into hard martensite while the inner core remains softer and ductile. This combination provides a good balance between hardness and torsional toughness [5].

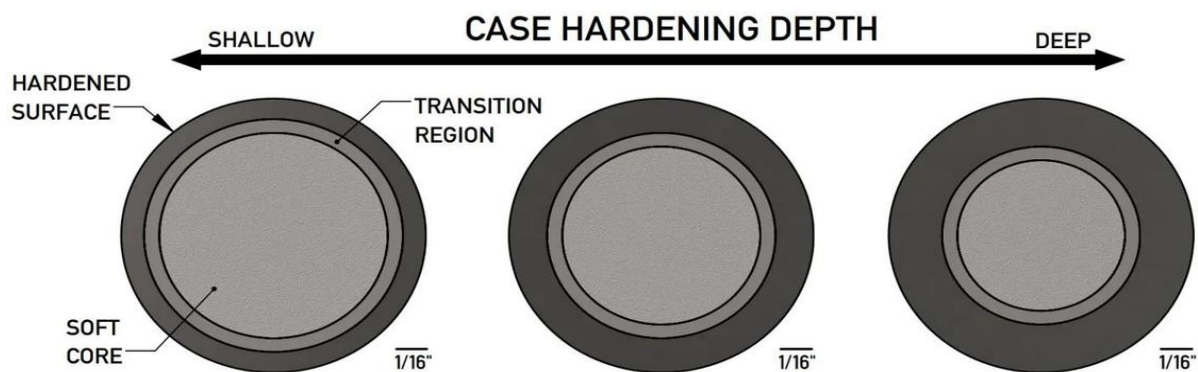


Figure 59

(c) Design Improvements

Additional design improvements include:

- Adding fillets at stepped regions to reduce stress concentration
- Improving surface finish to reduce vibration and friction
- Optimizing worm geometry for smoother meshing

These improvements can increase operational life and improve overall reliability of the hand mixer shaft.

6 Assembly

6.1 Assembling the components of the hand mixer

Fastners / Components	Images	Function in assembly
Self Tapping Blunt Roundhead Screw		Holds the top and the bottom parts of the housing together.
Self Tapping Screws		Fasteners used to hold in place motor and zinc chassis
Washers		Allows for movement of the gears
Clip		Holds Gear in place
Bolt		Holds all the parts of the motor housing together
Bolt		Holds the motor housing, stator and bottom part housing
Washer, Spring washer, Nut		Nut to keep above bolt secure, washers to allow for movement and keep tight.

6.2 3D Assembly exploded view

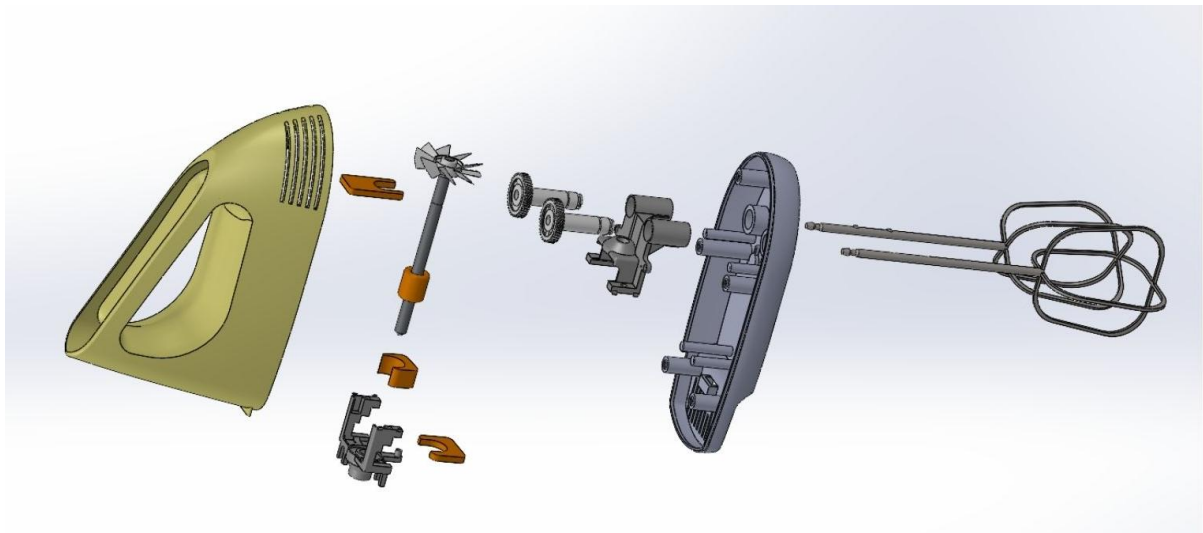


Figure 89 – Exploded view of assembly*

*Orange coloured components – bluff bodies to complete the assembly

6.3 Final assembly



Figure 90 – Assembled hand beater

7 References

- [1] S. Kalpakjian and S. Schmid, *Manufacturing Engineering and Technology*, 7th ed. Pearson Education, 2014.
- [2] R. K. Rajput, *A Textbook of Manufacturing Technology: Manufacturing Processes*. New Delhi: Laxmi Publications, 2007.
- [3] W. K. K. Lee, “Handheld Electric Mixer,” U.S. Patent US8162530B2, Apr. 24, 2012.
- [4] F. Mohamed and A. Nawawi, “Cost Reduction and Design Improvement of Hand Mixer for Baking Using Design for Manufacture and Assembly (DFMA),” *Progress in Engineering Application and Technology*, vol. 3, no. 1, pp. 672–682, 2022.
- [5] Nylon Technical Data Sheet, Quadrant Engineering Plastic Products, 2023.
- [6] J. A. Brooks, S. H. Goods, and C. V. Robino, “Weld Properties of a Free Machining Stainless Steel,” Technical Report, 2000.
- [7] G. Mundar et al., “Rapid Assessment of Steel Machinability through Spark Analysis and Data-Mining Techniques,” 2024.
- [8] M. C. Sadino-Riquelme et al., “Investigating a Stirred Bioreactor: Impact of Evolving Fermentation Broth Pseudoplastic Rheology on Mixing Mechanisms,” *Fermentation*, vol. 8, no. 3, 2022. doi: 10.3390/fermentation8030102.
- [9] S. Sarafan et al., “Benchmarking of 316L Stainless Steel Manufactured by a Hybrid Additive/Subtractive Technology,” 2022.
- [10] S. Zaki, N. Zhang, and M. D. Gilchrist, “Electropolishing and Shaping of Micro-Scale Metallic Features,” 2022.
- [11] Xometry, “Shore D Hardness Scale: Definition, Use, Types, Benefits, and Limitations.” [Online]. Available: <https://www.xometry.com/resources/materials/shore-d-hardness-scale/>. [Accessed: May 04, 2026].

- [12] Smooth-On, “Durometer Shore Hardness Scale.” [Online]. Available: <https://www.smooth-on.com/page/durometer-shore-hardness-scale/>. [Accessed: May 04, 2026].
- [13] QUANDA Plastic, “POM vs. Nylon: Performance Comparison and Selection Guide for Engineering Plastics.” [Online]. Available: <https://www.quandaplastic.com/pom-vs-nylon-performance-comparison-and-selection-guide-for-engineering-plastics/>. [Accessed: May 04, 2026].
- [14] Qianjiesi, “POM vs Nylon: Cost Comparison for Manufacturers – Beeplastic.” [Online]. Available: <https://www.beeplastic.com/blogs/plastic-insights/pom-vs-nylon-a-cost-analysis-for-manufacturers>. [Accessed: May 05, 2026].
- [15] Geomiq, “The Injection Moulding Process: A Complete Guide.” [Online]. Available: <https://geomiq.com/injection-moulding-guide/>. [Accessed: May 03, 2026].
- [16] Xometry Pro, “Injection Molding Surface Finishes: SPI and VDI.” [Online]. Available: <https://xometry.pro/en/articles/injection-molding-finishes/>. [Accessed: May 03, 2026].
- [17] STANMECH Technologies Inc., “Plastics Identification Flow Chart.” [Online]. Available: <https://www.stanmech.com/articles/plastics-identification-flow-chart>. [Accessed: May 03, 2026].
- [18] MatWeb, “Overview of Materials for Polycarbonate/ABS Alloy, Unreinforced.” [Online]. Available: <https://www.matweb.com/search/DataSheet.aspx?MatGUID=d16a64a389df424f8e53ed481e77477b&ckck=1>. [Accessed: May 03, 2026].
- [19] MatWeb, “Overview of Materials for Nylon 6, 30% Glass Fiber Filled.” [Online]. Available: <https://www.matweb.com/search/DataSheet.aspx?MatGUID=85375fdc53e64e3fbb59cac5516a69eb>. [Accessed: May 03, 2026].
- [20] S. Brumi, “3D Printing vs Injection Moulding vs CNC Machining,” 3D People UK. [Online]. Available: <https://www.3dpeople.uk/3dpeopleuk-blog/3dp-im-cnc/>. [Accessed: May 04, 2026].

- [21] L. DGMF Mold Clamps Co., “What is the Difference Between Two-Platen Mold and Three-Platen Mold?” [Online]. Available: <https://dgmfmoldclamps.com/what-is-the-difference-between-two-platen-mold-and-three-platen-mold/>. [Accessed: May 05, 2026].
- [22] Laszeray Technology, LLC, “A Comprehensive Look at ABS in Plastic Injection Molding.” [Online]. Available: <https://laszeray.com/a-comprehensive-look-at-abs-in-plastic-injection-molding/>. [Accessed: May 11, 2026].
- [23] F. Peng, “ABS Injection Molding: Engineering Guide, Process Control, and Applications,” RapidDirect, Feb. 26, 2022. [Online]. Available: <https://www.rapiddirect.com/blog/abs-injection-molding/>.
- [24] AAA MOULD, “Auto-Unscrewing Moulds: Complete Technical Guide,” Oct. 16, 2025. [Online]. Available: <https://www.aaamould.com/news/auto-unscrewing-moulds-complete-technical-guide.html>.
- [25] J. Zhu, Z. Qiu, and W. Huang, “Development and Application of High-Gloss Injection Molding Technology,” *Journal of Physics: Conference Series*, vol. 2206, no. 1, p. 012016, 2022.
- [26] R. Zhao, X. Li, Z. Wang, T. Wu, and J. Liang, “Effect of Injection Molding Parameters on the Tensile Strength of Short-Carbon-Fiber-Reinforced Nylon 6,” *Polymers*, vol. 17, no. 9, p. 1264, May 2025.
- [27] S. Wang, Y. Yan, S. Jiang, and F. Wang, “Surface Reconstruction Based on Reverse Engineering and Testing,” in *Proc. 2nd Int. Conf. Modelling, Identification and Control (MIC)*, 2015, pp. 107–112.
- [28] TECLOCK, “Durometer & IRHD Hardness Tester,” Teclock Corporation, Catalog, 2026.
- [29] Omni Materials, “Shore D Hardness Test (Durometer Scale) - Hardness of Plastic Materials.” [Online].
- [30] Boedeker Plastics, Inc., “Burn Test Chart for Plastic Identification,” Technical Data.